Cost Uncertainty in Sequential Strategy Validation: A Diagnostic Study with Adaptive Bitcoin Trend Following

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Abstract

Sequential strategy comparisons can lose evidence through two distinct forms of cost pessimism: bounding the two portfolios separately, and allowing one stationary cost coefficient to vary adversarially across periods. We quantify these effects using established bounded-betting and empirical-Bernstein methods. Two separately specified synthetic experiments contain 24 and 25 cases, with 4,000 independent worlds within each case. Matched comparisons distinguish cost geometry, evidence form, volatility adaptation and whole-history nuisance profiling. At effect 0.05, under high volatility, alternating turnover contrasts and fixed cost information, coherent empirical-Bernstein power is 33.500% versus 0.000% for local bounds. Adaptive betting does not uniformly improve power, and a symmetric upper-cost shortcut can be anti-conservative when the reference trades more. These findings are conditional on a bounded utility model and simultaneous calibration coverage; they do not validate a market execution-cost model. A separate audited, retrospective Bitcoin application finds no evaluation-window switches by the four-hour adaptive selector: its 2.607% return equals the fixed EMA 50/200 baseline. Hourly and daily adaptive differences have negative point estimates and broad approximate intervals. The contribution is a reproducible diagnosis of information loss and invalid shortcuts, together with a negative trading result. No new general theorem or demonstrated adaptive market edge is claimed.

1 Introduction

An adaptive trading rule can appear attractive for several different reasons: it may select useful information, change its market exposure, trade less, or benefit from an optimistic execution assumption. Repeatedly checking its performance introduces another distinction: an attractive historical difference does not automatically constitute valid sequential evidence. This study examines those distinctions for a small, transparent family of Bitcoin trend rules.

The statistical question concerns a paired utility difference after economic materiality and uncertain costs. If strategy and reference pay a defensibly shared mean cost, bounding their costs separately discards information. If the reference trades more, charging both sides the same upper cost can instead produce an optimistic difference. A valid analysis must distinguish those geometric effects from the choice of evidence process, uncertainty about the cost model, and the trading performance itself.

Our contribution is a diagnostic experimental study, not a new general theory of betting or robust optimization. One frozen study diagnoses shared-cost geometry; a second isolates evidence form, volatility adaptation and preservation of a common cost parameter across time. We implement matched constructions, audit explicit failure cases, and preserve independent-world outcomes and all research attempts. A separate retrospective Bitcoin application uses exact self-financing books, inherited holdings, matched dates, simple controls and assumed-cost

sensitivities. The bounded synthetic guarantee is not transferred to market P&L. This separation makes favorable and unfavorable findings equally interpretable.

Our first experiment asks how much evidence is discarded when a paired comparison ignores a defensibly shared cost coefficient. For example, if both turnovers equal one and the coefficient lies in $[0, 0.2]$, the paired cost difference is exactly zero, whereas separate portfolio bounds subtract 0.2 from the lower utility difference. This is an elementary identity; the experiment measures its consequences for finite-horizon detection and false discoveries. Our second experiment asks how much further evidence is lost by minimizing each period independently when the unknown coefficient is actually constant across time. It compares matched local and whole-history constructions and separates this operation from changing the betting rule. A final Bitcoin case study examines what the implemented trading comparison establishes when those synthetic calibration assumptions are unavailable. It is an application audit, not a transfer of the synthetic guarantee to market returns.

1.1 Related work and scope of the contribution

Choe and Ramdas [1] develop sequential paired forecast comparisons, including an average conditional-score null. Arnold et al. [2] study sequential model confidence sets. Bounded betting and time-uniform concentration are established tools [3, 4]; online e-value procedures already address multiple testing [5, 6]. Our chosen error criterion is family-wise error, not false discovery rate. Neither criterion repairs an invalid input process. Li and Zhu [17] project simultaneous uncertainty regions onto a constrained conformal-pipeline optimizer. Their coverage-feasibility target differs from our paired execution-cost utility contrast; broad uncertainty-set selection language is therefore not itself a novelty claim here.

Financial sequential validation also has close predecessors. Jiao et al. [7] compare risk forecasts with e-backtests. The indexed abstracts of Liu [8] and Stephan [9] describe, respectively, sequential factor-alpha testing and Bayesian economic activation of alpha signals. Their full SSRN manuscripts were inaccessible in this review, so we make no detailed theorem comparison or priority claim against them.

Whole-history nuisance profiling is already present in universal inference [10]; composite e-processes may be useful even when a common nonnegative supermartingale construction is powerless [11]. Consequently, preserving one cost parameter across time is not by itself a new principle. Worst-case linear cost contrasts are elementary robust counterparts [12]. Our comparison isolates the consequences of applying these established ideas correctly or incorrectly.

Fixed-sample time-series comparisons already address model search and data reuse, including White's Reality Check [18] and the model confidence set of Hansen et al. [19]. Our bounded sequential experiment uses a different conditional information model; the market application does not implement those procedures or claim selection-adjusted superiority.

Crypto trend-following comparisons with trading costs predate this study [13]. Bysik and Ślepaczuk [14] evaluate hourly Bitcoin machine-learning strategies with walk-forward selection, cost-aware execution and dependent-data resampling. We do not claim that adding costs, Bitcoin, or a chronological evaluation is a novel method. Our application instead makes the connection between validity assumptions, data outages, simple adaptation controls and exact cost accounting explicit. Cross-venue observations remain observations of the same asset.

The nearest-work register records inspected passages and access limitations. It does not certify exhaustive priority. In particular, no claim of a first sequential trading test, first cost-aware Bitcoin strategy, or newly discovered profitable trading opportunity is made.

Evidence family	Cost treatment	Supported null	Scope of inference
Fixed linear / Hoeffding mixtures	Oracle, local joint, local separate	Prefix-weak; hence strong	Bounded synthetic model; fixed component grids and weights
Fixed empirical-Bernstein mixtures	Oracle, local joint, local separate, coherent joint profile	Prefix-weak; hence strong	Predictable common centers; robust minima over full utility intervals
One-sided aGRAPA specialization	Oracle, local joint, local separate	Strong conditional only	Predictable adaptive stakes; no prefix-weak guarantee asserted
Historical market comparison	Assumed cost scenarios with exact cash/coin books	No sequential null tested	Approximate paired moving-block intervals; retrospective and unadjusted

Table 1: Distinct guarantees and applications. Synthetic cost-based guarantees require the bounded support, oracle information, predictability, fixed shared coefficient and simultaneous calibration assumptions in the text. The family bound is $0.04 + 0.01 = 0.05$ separately within a comparator; pooling discoveries across comparators has no inherited 5% guarantee. Oracle thresholds are kept matched for comparison. Market intervals do not inherit synthetic guarantees.

2 Statistical setup and established-method construction

2.1 Information, utility and null hypotheses

For a candidate born at a predictable research time, index its subsequent non-overlapping trials by i. Candidate choice, turnover coefficients $q_i^A, q_i^R \geq 0$, materiality m_i, cost interval and betting components are fixed before the affected outcome is observed. The outcome matures after its issue time. No intermediate label is available in the delayed experiment.

Let G_i denote the observed gross utility difference and θ a fixed unknown common mean-cost coefficient under each generating law. Define

$$d_i = q_i^A - q_i^R, \qquad D_i = G_i - d_i\theta - m_i. \tag{1}$$

The synthetic model assumes a known $B > 0$ with $|D_i| \leq B$ almost surely. These are bounded experimental utilities, not empirical bounds inferred from the largest historical return. A common expected cost need not mean equal realized slippage on every trade.

Write $\mathcal{B}_i$ for the pretrial oracle information, including actual researcher information, calibration, policy choices, and all previously matured latent utilities relevant to the process. Put $\mu_i = \mathbb{E}[D_i \mid \mathcal{B}_i]$. The strong null is $\mu_i \leq 0$ at every post-birth trial. The prefix-weak null is

$$\sum_{i=1}^{n} \mu_i \leq 0 \quad \text{almost surely for every } n. \tag{2}$$

The latter permits temporary positive conditional means after an initial negative period. All prefixes begin at the candidate's declared birth. A null defined only using a smaller observable history is not interchangeable with this oracle null. Information timing matters for delayed evidence as well [15].

2.2 One simultaneous cost envelope

Calibration observations $C_k \in [0, c]$ obey $\mathbb{E}[C_k \mid \text{full prior calibration/game information}] = \theta$. At $k \geq 1$, let $\bar{C}_k$ be their average and set

$$\delta_k = \frac{6\delta}{\pi^2 k^2}, \qquad r_k = c\sqrt{\frac{\log(2/\delta_k)}{2k}}, \tag{3}$$

$$[\ell_k, u_k] = [\max(0, \bar{C}_k - r_k), \min(c, \bar{C}_k + r_k)]. \tag{4}$$

At $k = 0$ use $[0, c]$. Conditional bounded-difference Hoeffding concentration and a union bound imply

$$\mathbb{P}(\mathcal{G}^c) \leq \delta, \quad \mathcal{G} = \{\theta \in [\ell_k, u_k] \text{ for all } k\}. \tag{5}$$

This deliberately transparent spending envelope can be conservative. One global shared envelope is charged once. Unrelated cost parameters cannot silently reuse its error allowance. Bounds concern the fixed mean coefficient, not next-trade realized costs or a permanently nonstationary slippage process.

At issue time use the latest available interval $[\ell_i, u_i]$ to form

$$L_i^J = G_i - \max(d_i \ell_i, d_i u_i) - m_i, \tag{6}$$

$$L_i^S = G_i - q_i^A u_i + q_i^R \ell_i - m_i. \tag{7}$$

Both are at most D_i on $\mathcal{G}$. Their exact width penalty is

$$L_i^J - L_i^S = \min(q_i^A, q_i^R)(u_i - \ell_i). \tag{8}$$

This is an elementary identity, not a new robust-optimization result. When turnovers match, the common coefficient cancels in the paired contrast, whereas the separate envelope still pays its uncertainty width. Charging u_i to both sides gives $G_i - d_i u_i - m_i$, which is optimistic when $d_i < 0$.

Define $Z_i = \text{clip}(L_i, -B, B)$. On $\mathcal{G}$, $Z_i \leq D_i$: clipping below $-B$ is safe because $D_i \geq -B$, and the upper endpoint cannot exceed D_i on that event. This does not justify clipping financial returns while retaining an unmodified economic estimand.

2.3 Fixed mixtures and error composition

For fixed components $\lambda_k \in [0, 1)$ and nonnegative fixed weights summing to one, the linear proxy mixture is

$$W_n = \sum_k \pi_k \prod_{i=1}^{n} (1 + \lambda_k Z_i / B). \tag{9}$$

The matched Hoeffding mixture replaces each product by $\exp\{\lambda_k \sum_{i=1}^n Z_i/B - n\lambda_k^2/2\}$. Components and weights are fixed at birth; independently refitting stakes does not inherit the weak-null argument below.

Proposition 1 (Standard domination and cost-error composition)**.** *Under the stated boundedness, oracle-filtration, calibration and prefix-weak null assumptions, let stream j receive a predictable error allowance α_j at its birth, with $\sum_j \alpha_j \leq \alpha_{\text{test}}$ pathwise. Reject it when its proxy mixture reaches $1/\alpha_j$. Then*

$$\mathbb{P}(\textit{at least one true-null rejection}) \leq \alpha_{\text{test}} + \delta.$$

The result also holds for the strong null, a subset of the prefix-weak null.

Proof. For a fixed component at the true coefficient, define

$$M_{k,n} = \prod_{i=1}^{n}(1 + \lambda_k D_i/B), \qquad N_{k,n} = \prod_{i=1}^{n} \frac{1 + \lambda_k D_i/B}{1 + \lambda_k \mu_i/B}.$$

The denominator is positive and predictable. The ratio product is a nonnegative martingale starting at one in the oracle history. By $1+x \le e^x$ and (2), $\prod_i(1+\lambda_k\mu_i/B) \le \exp\{\lambda_k \sum_i \mu_i/B\} \le 1$, hence $M_{k,n} \le N_{k,n}$. Their fixed mixture is therefore dominated by a test martingale. For the Hoeffding form, the centered conditional Hoeffding exponential is a test supermartingale; its conditional-mean compensator is also at most one under (2).

On the single event $\mathcal{G}$, the monotonicity of every factor in Z_i gives proxy wealth at most the corresponding latent wealth, simultaneously over all evaluated times and streams. Thus a proxy crossing on $\mathcal{G}$ implies a latent crossing. Conditional Ville bounds at predictable births and a union bound give total latent true-null crossing probability at most $\mathbb{E}\sum_j \alpha_j \le \alpha_{\text{test}}$. Adding $\mathbb{P}(\mathcal{G}^c) \le \delta$ proves the assertion. We do not condition the data-generating law on $\mathcal{G}$, and do not assert that the proxy process itself is a supermartingale. □

This proposition collects established bounded-betting, compensator and coverage-union arguments for the implemented diagnostic; it is not claimed as a new general theorem. Under the strong null, predictable changing stakes also admit the usual latent supermartingale argument, but that is not the registered fixed-mixture comparison. The registered family budget is $0.05 = 0.04 + 0.01$. Fixed families use $\alpha_j = 0.04/H$; sequential births use $\alpha_j = 0.04/[j(j+1)]$. Candidate dependence does not require an additional independence assumption. Past prescreen labels are never reused as post-birth evidence.

For identical component grids, weights and streams, (8) implies pathwise joint wealth at least separate wealth. Consequently a joint-only first-stream rejection is a binary paired event whose Monte Carlo uncertainty is computed across worlds. The difference is called a rejection-rate difference under a null and a power difference under an alternative. This dominance claim does not compare different betting rules or guarantee higher investment returns.

2.4 Variance-adaptive comparators and one coefficient across time

The follow-up uses established alternatives to the inactive Hoeffding comparators. Normalize the bounded utilities to $B = 1$ and write $A_i = G_i - m_i$, so that $D_i(\vartheta) = A_i - d_i\vartheta$. For predictable $\gamma_i \in [-1, 1]$, fixed $0 \le \lambda < 1/2$ and

$$\psi(\lambda) = \frac{-\log(1-2\lambda) - 2\lambda}{4}, \qquad f_i(\vartheta;\lambda) = \lambda D_i(\vartheta) - \psi(\lambda)\{D_i(\vartheta) - \gamma_i\}^2,$$

the true-parameter exponential has the empirical-Bernstein domination of Choe and Ramdas [1]. The centered process

$$N_n(\lambda) = \exp\left\{\sum_{i=1}^{n}[\lambda(D_i - \mu_i) - \psi(\lambda)(D_i - \gamma_i)^2]\right\}$$

is a test supermartingale. Indeed, $D_i - \gamma_i \ge -2$ and the bounded-score inequality gives $\exp\{\lambda(D_i - \gamma_i) - \psi(\lambda)(D_i - \gamma_i)^2\} \le 1 + \lambda(D_i - \gamma_i)$. After conditional expectation and multiplication by $\exp\{\lambda(\gamma_i - \mu_i)\}$, the right side is $e^{-s}(1+s) \le 1$, where $s = \lambda(\mu_i - \gamma_i) > -1$. Under (2), the uncentered true-parameter product is no larger than N_n. Fixed mixtures therefore retain prefix-weak validity.

The factor is not monotone in utility. For the local robust construction, minimize f_i over the entire joint utility interval induced by the cost set, or over the larger separate interval

$$[A_i - q_i^A u_i + q_i^R \ell_i, A_i - q_i^A \ell_i + q_i^R u_i].$$

Concavity makes the minimum the smaller endpoint value. Simply substituting a lower utility into an empirical-Bernstein factor can be optimistic.

Let $C_n = [\ell_n, u_n]$ be the nonempty running intersection of calibration intervals available when label n was issued. The componentwise coherent profile is

$$W_n^{\mathrm{coh}} = \sum_k \pi_k \exp\left\{ \inf_{\vartheta \in C_n} \sum_{i=1}^{n} f_i(\vartheta; \lambda_k) \right\}. \tag{10}$$

The parameter remains the same throughout each historical product. This is the established composite-inference principle, specialized to the paired cost model [10, 11]. On simultaneous coverage, each profiled component is at most its value at the true coefficient and thus its latent dominating process. The event-inclusion proof of Proposition 1 applies with the same additive coverage budget. It does not condition on coverage or assert that the observed profile itself is a supermartingale.

The sum of log factors is a concave quadratic in ϑ. Retain

$$S_A = \sum A_i, \quad S_d = \sum d_i, \quad Q_{AA} = \sum (A_i - \gamma_i)^2, \quad Q_{Ad} = \sum (A_i - \gamma_i) d_i, \quad Q_{dd} = \sum d_i^2.$$

Then evaluate

$$\lambda(S_A - \vartheta S_d) - \psi(\lambda)(Q_{AA} - 2\vartheta Q_{Ad} + \vartheta^2 Q_{dd})$$

at the two current endpoints and take the smaller value for each component. These five sums have constant state in the number of labels. Minimizing the whole mixture only at endpoints would need an additional shape argument and is not the implemented procedure.

For a matched local construction, $C_n \subseteq C_i$ implies

$$\inf_{\vartheta \in C_n} \sum_{i=1}^{n} f_i(\vartheta; \lambda) \geq \sum_{i=1}^{n} \inf_{\vartheta \in C_i} f_i(\vartheta; \lambda).$$

Exponentiating and mixing gives coherent evidence at least local joint evidence. Local joint evidence likewise dominates the matched separate interval construction. These are comparisons within the same evidence form, centers and grid. They do not order different test families or subsequent deployment utility. Empty cost intersections permanently suppress new cost-based crossings, rather than being assigned infinite evidence; prior discoveries and their specified paper allocation are retained.

The additional adaptive linear comparator is a declared one-sided aGRAPA specialization [3, Appendix B.3]:

$$\lambda_i = \mathrm{clip}\left(\frac{\sum_{r<i} x_r}{\sum_{r<i} x_r^2}, 0, 0.8 \right).$$

Its initial stake is 0.1; a later zero denominator gives stake zero. Each oracle/joint/separate variant fits its own past utilities. Predictability and bounded stakes give strong conditional-null validity by the ordinary linear-factor argument, with latent domination for cost proxies. Arbitrary adaptive stakes do not inherit the prefix-weak proof. For example, 400 deterministic utilities -0.06 followed by 400 utilities $+0.06$ satisfy every prefix-weak null. Stakes zero in the first half and 0.8 in the second produce wealth $(1 + 0.8 \cdot 0.06)^{400} > 250$. No joint/oracle pathwise ordering is claimed for separately refitted stakes.

Within a case, the family error bound is applied separately to each comparator's ten streams. Pooling discoveries across all comparator methods, or choosing a favorable method after viewing the comparison, does not inherit a 5% family-wise bound.

2.5 The market application is a different experiment

Actual self-financing books contain nonlinear cost/wealth interactions, band-triggered trades and different inherited positions. They are not assumed to obey (1) with a known bounded latent difference. Their costs are scenario assumptions, without calibration observations satisfying (5). We therefore report paired historical log differences and approximate dependent-data intervals, without marketing them as sequential discoveries. This restriction is part of the research design, not a correction chosen after seeing its profitability.

3 Experimental design

3.1 Paired cost-geometry experiment (v1)

The locally registered v1 specification fixes 24 cases, 4,000 independent worlds per case, 800 calendar steps, deterministic case-specific seeds, $B = 1$, $\theta = 0.08$, calibration support $[0, 0.2]$, and materiality $m = 0.01$ in synthetic units. The fixed grid is $\lambda \in \{0.05, 0.1, 0.2, 0.4, 0.8\}$ with equal weights. Default families have ten streams; search-size cases have thirty. Five candidates are introduced adaptively in the birth cases. All cases, including invalid-model stress cases, are retained. Separate 128-world, 60-step smoke tests are implementation checks and are excluded from substantive estimates.

At issue time t, let the previously matured public common sign be S_{t-1}. With fresh common and candidate-specific independent Rademacher signs S, I_j,

$$\sigma_t = 0.2 + 0.1\{1 + \sin(2\pi t/80)\} + 0.15\mathbf{1}\{S_{t-1} > 0\}, \tag{11}$$

$$\epsilon_{j,t} = \sigma_t(0.6S + 0.4I_j), \tag{12}$$

$$q_t^A = 0.7 + 0.1\mathbf{1}\{S_{t-1} > 0\}, \qquad q_t^R = \rho q_t^A. \tag{13}$$

The fresh common sign is publicly observed at maturity. The reference-more- turnover cases exchange the two coefficients. Generate latent materiality- adjusted utility $D_{j,t} = \epsilon_{j,t} + a_j$ and observed $G_{j,t} = D_{j,t} + (q_t^A - q_t^R)\theta + m$. Even-indexed candidates receive the case's positive effect and odd-indexed candidates remain null. Because D already subtracts materiality, zero effect means zero advantage beyond the hurdle, not zero unadjusted net utility. The weak-prefix case uses alternating conditional effects $-0.07, +0.07$, starting negative. Thus the assumed bound holds by construction while volatility, dependence and conditional means need not be constant.

Default initial calibration is 32 independent scaled Bernoulli observations, represented by their sufficient sum; zero- and 512-sample cases vary this budget. Add one calibration observation after each maturity. An observation from the current trial cannot enter its issue-time interval. One declared violation shifts the true coefficient to 0.18 halfway through the horizon, outside the stationary-mean calibration assumption.

Adaptive births choose the highest cumulative past plugin utility among unused pool candidates, with deterministic index tie-breaking. Only future post-birth trials enter their evidence. Delayed trials issue at $0, d+1, 2(d+1), \ldots$ and mature at issue plus $d+1$. There is no intermediate feedback, so delay reduces both matured outcomes and new calibration samples over a fixed calendar horizon. This does not establish a general overlapping-label theorem.

Comparators are the known-cost oracle, joint and separate linear mixtures, plugin and symmetric-upper linear mixtures, joint and separate Hoeffding mixtures, a fixed-horizon Hoeffding boundary evaluated only at the horizon, and the same fixed-time boundary incorrectly peeked at repeatedly. Oracle tests use the same 0.04 test budget for matched comparison. Plugin and symmetric-upper comparators have no asserted guarantee.

Primary summaries are true-null family-wise rejection and true-alternative power with world-level Wilson intervals. Detection times use calendar time from origin; nonrejections are horizon-censored, and detection-conditional medians are labeled. Paper allocation activates the lowest-index first discovered candidate on the following trial and accumulates additive utility. It has no

funding, compounding, market orders or realized execution-cost shocks. Normal Monte Carlo intervals are unavailable for degenerate samples rather than reported as falsely certain $[0, 0]$. Runtime and stored world arrays accompany every case.

3.2 Evidence form and temporal coherence experiment (v2)

After all v1 outcomes and the market study were inspected, a separate v2 protocol was written, independently reviewed and frozen before its substantive run. The inactive Hoeffding comparators and fixed-sign turnover contrasts in v1 motivated the follow-up. Its specification is not a retroactive amendment to the baseline or a new market holdout.

The 24-cell factorial crosses effect $a \in \{0, 0.02, 0.05\}$, volatility scale $v \in \{1, 0.4\}$, positive versus alternating turnover difference, and static initial versus contracting cost information. Every case has 4,000 worlds, 800 non-overlapping labels and ten simultaneous streams; only stream zero receives a positive effect. The separate prefix-weak diagnostic gives every stream conditional mean -0.06 for 400 labels and $+0.06$ for 400 labels. All streams in that diagnostic are prefix-weak nulls; adaptive-bet methods are explicitly labeled unsupported for that null.

Use the v1 common/idiosyncratic fair-sign construction, multiplied by v. Turnovers are $(q^A, q^R) = (0.8, 0.4)$; alternating cases swap them every 20 labels, beginning with positive difference. Generate $D_{j,t} = a_j + v\sigma_t(0.6S_t + 0.4I_{j,t})$ and $G_{j,t} = D_{j,t} + d_t\theta + m$, with $\theta = 0.08$, $m = 0.01$. The deterministic support bound, including all separate-cost endpoints, is at most $0.65 + 0.06 + 0.4 \cdot 0.08 + 0.8 \cdot 0.2 = 0.902 < 1$. The recorded sample maximum is only a diagnostic of this bound.

Initialize calibration with 32 independent observations equal to 0.2 with probability 0.4 and zero otherwise. Static cases retain the initial interval. Contracting cases intersect all successive intervals, with one new sample revealed after each evidence update. The last post-horizon sample cannot affect a test; its conflict flag is recorded separately. The inherited k^{-2} calibration spending is conservative: querying only $k \geq 32$ gives union bound $\delta(6/\pi^2)\sum_{k\geq 32} k^{-2} \simeq 0.000193$, below the unchanged reserved 0.01. This design does not optimize calibration efficiency.

The ten comparators comprise three fixed linear mixtures and three aGRAPA variants, each with oracle, joint and separate costs; oracle EB, local joint EB and local separate EB; and the coherent joint EB profile. The equally weighted fixed grids are

$$\Lambda_{\text{linear}} = \{0.05, 0.1, 0.2, 0.4, 0.8\}, \qquad \Lambda_{\text{EB}} = \{0.025, 0.05, 0.1, 0.2, 0.4\},$$

with $c = 2$ for EB. The shared predictable EB center is the mean of prior observed plugin utilities, clipped to $[-1, 1]$, initially zero. Each past plugin value uses the coefficient sample mean available when its label was issued; old centers are never refitted. Cost-information comparisons also change center quality and adaptive stakes, so only matched within-cell profiling comparisons isolate that operation cleanly.

Each method receives the same 0.04 test budget over its ten streams, reserving 0.01 for the shared cost event. Oracle thresholds remain matched. The three primary descriptive contrasts are joint aGRAPA minus joint fixed linear, coherent EB minus joint local EB, and joint local EB minus separate local EB at effect 0.05. Every cell, including zero and weak effects, is retained. There is no prespecified universal ordering between base method families.

Separate outcome and calibration random streams use seed 20261023 and align innovations by world index across factorial cells. Worlds are independent; streams and factorial cells are not independent replications. Report Wilson intervals for binary events and paired world-level normal Monte Carlo intervals for signed differences, horizon-censored time and additive materiality-adjusted utility. They are estimation summaries, not simultaneous discovery guarantees. Activation begins only on the label after first discovery, with the lowest index breaking ties. Runtime and named persistent state bytes are recorded; neither is a controlled hardware benchmark or process peak-memory measurement. This is a bounded diagnostic experiment, not a calibrated Bitcoin execution or return model.

3.3 Audited retrospective Bitcoin application

The separate market protocol fixes Binance BTC/USDT spot four-hour evaluation from 1 January 2025 through 20 October 2026. A second, common window from 20 February through 20 October 2026 compares native one-hour, four-hour and daily candles. Its start is determined by coverage and the strict selector history requirement before new-period P&L was evaluated. Bitstamp BTC/USD four-hour candles repeat both windows. These six cases are predetermined, not ranked to choose a final strategy.

Binance raw archives retain vendor checksums and immutable local hashes. The importer normalizes the documented microsecond spot timestamp change beginning in 2025. Independent standard-library reconciliation checks every retained OHLCV observation, excluded-row hash and source boundary. The secondary Bitstamp snapshots have local response hashes but no independently published vendor checksum; close timestamps are nominal interval endpoints inferred from its opening timestamps. No source certifies point-in-time availability against later historical revisions.

The reference is an EMA 50/200 long/cash rule. The adaptive family is EMA 25/100, 50/200 and 100/400. At four-hour resolution, risk targets 20% annual volatility using 180 prior log returns, capped at unit unlevered weight. Features require 2,000 bars of continuous warm-up. A 4,380-return trailing window of continuous net shadow-book returns is ranked every 180 bars; ties retain the incumbent and then prefer the reference. Clock durations are preserved under timeframe conversion with upward integer rounding.

Orders use the previous completed close's target and the next observed open. The absolute weight rebalancing band is 0.10. A missing interval cancels crossing orders and resets features while spot holdings are carried. The incumbent can resume after feature warm-up; a new selection waits for full selector history. Cash earns zero. Fees apply to traded notional and exact post-fee target weights are solved, not approximated. Books inherit pre-window positions and terminal holdings are marked, with a separately labeled estimated liquidation sensitivity.

All nine reported policies are fixed, adaptive, cash, buy-and-hold, volatility- targeted BTC, slow reference resizing, equal target-weight ensemble, and the two additional fixed constituents. The ensemble includes the reference. Slow resizing delays adjustment of existing nonzero exposure but still executes entry/exit transitions. Assumed all-in costs are 0, 10, 25, 50 and 100 basis points per traded side, with 25 as the base. Base-cost selection choices are frozen across cost sensitivity; portfolio accounting is rerun rather than subtracting a linear ex-post fee estimate.

The primary endpoint is adaptive-minus-fixed mean net log return, scaled by 2,190 bars/year, with materiality 0.02 annual log units. A non-circular paired moving-block percentile bootstrap uses 5,000 draws and a primary 30-day block, with 15- and 60-day sensitivity [16]. Fewer than five blocks or missing scored intervals makes that interval unavailable. Those are approximate retrospective summaries requiring adequate process stability and weak dependence. The five-block criterion is a reporting heuristic, not a theorem guaranteeing coverage. Secondary controls, costs, years and venues have unadjusted exploratory intervals. None establishes a research-selection-adjusted or sequential claim.

Every bar's log return is decomposed exactly into prior-close to pre-trade-open price movement, pre/post-trade fee drag, and post-trade-open to close holdings return. Their sum and fees are checked against wealth. Differences in those components are accounting descriptions, not causal selection alpha. The data cannot identify unmeasured latency, capacity, spread or actual fill quality.

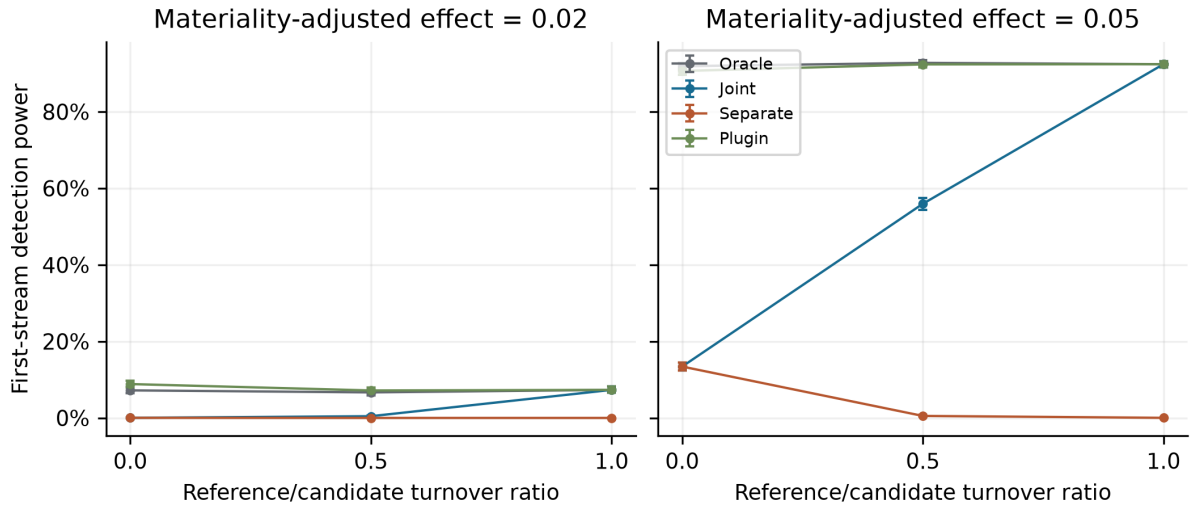


Figure 1: Overlap/effect comparison with 95% Wilson intervals over independent worlds. Methods within a case use paired data. Plugin is unguaranteed. Lines connect three evaluated points; overlap cases use separate seeds.

4 Results

4.1 Validity and discarded shared structure

All 24 registered cases completed at the substantive 4,000-world scale. The largest observed joint-linear true-null family rejection rate among the 23 valid-model cases is 0.675%, below the nominal 5% ceiling. This checks the implemented model; the proof remains conditional on Section 2. No calibration-envelope failure was observed in those cases. Zero out of 4,000 has a Wilson upper limit of 0.096% within each case, not zero failure probability or simultaneous coverage across cases. No stored crossing-inclusion violation was recorded.

Figure 1 compares matched methods on the same worlds. At effect 0.05 and equal turnover, joint first-stream power is 92.425% [91.564%, 93.205%], versus 0.025% [0.004%, 0.141%] for separate bounds. The paired improvement is 92.400 percentage points [91.537, 93.181], using a Wilson interval for the joint-only rejection event. Joint and known-cost oracle results coincide because the common coefficient cancels exactly.

At half overlap, joint/separate powers are 55.950%/0.525%. At zero reference turnover their proxies coincide. At effect 0.02, even perfect cancellation gives only 7.325% joint power; at effect 0.005 it is 0.475%. With 512 initial calibration observations, half-overlap joint/separate powers rise to 74.925%/18.425%. Both calibration quality and uncertainty geometry matter. These scale-dependent results do not demonstrate generally powerful strategy discovery or estimate real trading fees.

The invalid symmetric-upper rule gives 12.950% false family rejection [11.945%, 14.026%] when the reference trades more (Figure 2). A conservative cost convention for each portfolio is therefore not necessarily conservative for their difference. Under the explicitly invalid cost-shift law, every world's envelope eventually fails and plugin false rejection is 43.550% [42.020%, 45.092%]. Low joint rejection in that particular violation case does not restore its missing stationarity guarantee.

All four Hoeffding, fixed-time and peeked controls have zero rejections in every case. Their support bound and the chosen signal/noise scale are too conservative to provide an informative empirical optional-stopping comparison. These results do not demonstrate peeking inflation, and absence of observed inflation does not validate peeking. Appendix B addresses separate invalidity mechanisms exactly, without substituting a favorable new simulation.

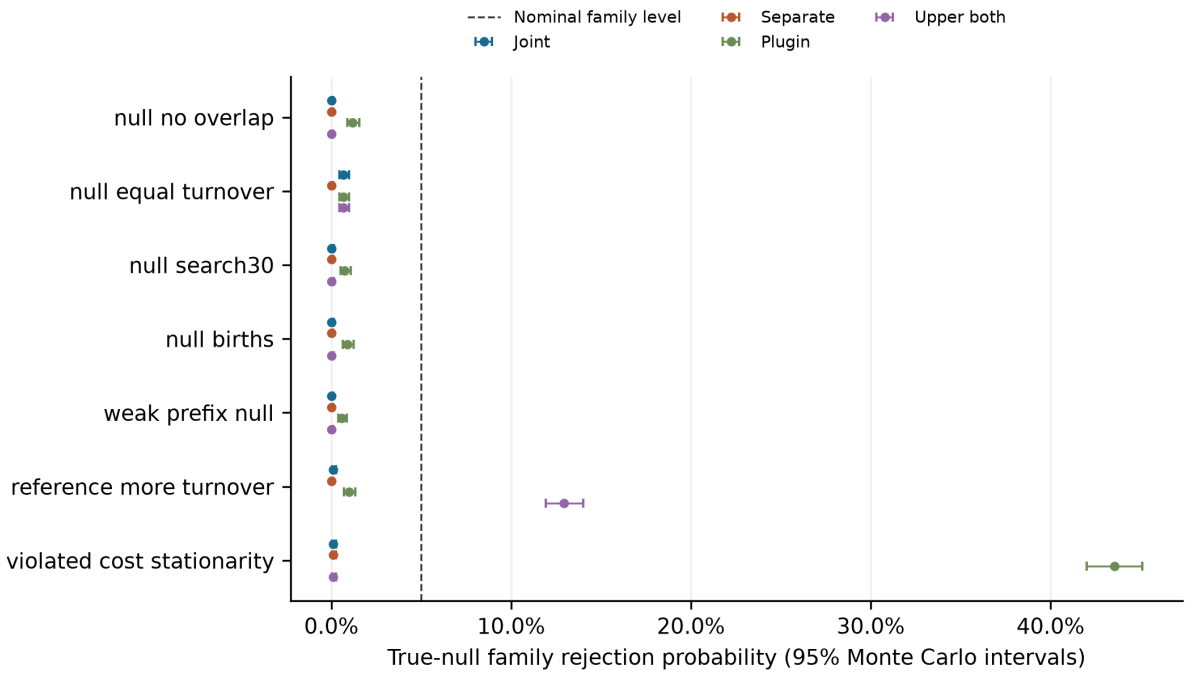


Figure 2: Selected null diagnostics; the supplement retains every method/case. The final row violates stationary calibration. Upper both and Plugin have no asserted guarantee. Zero estimates retain nonzero Wilson upper bounds. The dashed line is the declared 5% family level.

4.2 Timing, utility and computation

Equal-turnover joint restricted first-alternative detection time is 298.3 calendar steps versus 799.9 for separate bounds. The paired mean paper-utility difference is 24.946 synthetic units [24.650,25.242], an approximate normal Monte Carlo interval. This reflects the registered allocation rule and effect, not dollars, compounded wealth, actual execution quality or general optimality (Figure 3).

Delay reduces matured labels from 800 to 160 and calibration opportunities, so the lower power does not isolate delay alone. Birth cases change the family from ten streams to five, give the first birth a larger allowance (0.02 versus 0.004), and stagger observation counts. Higher first-stream power cannot be attributed solely to adaptation or prescreening. Within-case joint/separate contrasts remain paired.

Case runtimes sum to 2,715.6 seconds with concurrent local work; this is a recorded-host observation, not a controlled speed benchmark. Mixture-array bytes are reported separately from whole-process peak memory, which was not measured. Saved worlds support independent truth-mask, crossing-time, summary and activation checks. Per-trial latent trajectories require the frozen generator and seed; they are not stored in the final world arrays.

4.3 Separating evidence form from temporal cost pessimism

All 25 separately frozen follow-up cases completed at 4,000 worlds and 800 labels. Across method/case combinations within their stated null scope, the largest observed true-null family rejection rate is 1.150%. No used cost-envelope failure or interval conflict was observed. These frequencies support the implementation checks; they do not establish sharpness of the nominal 5% bound. The calibration spending rule is deliberately conservative, and neither cases nor comparators constitute one jointly controlled family of research conclusions.

Figure 4 displays all eight prespecified effect-0.05 factorial cells. With high volatility, fixed cost information and alternating turnover differences, coherent EB has 33.500% first-stream

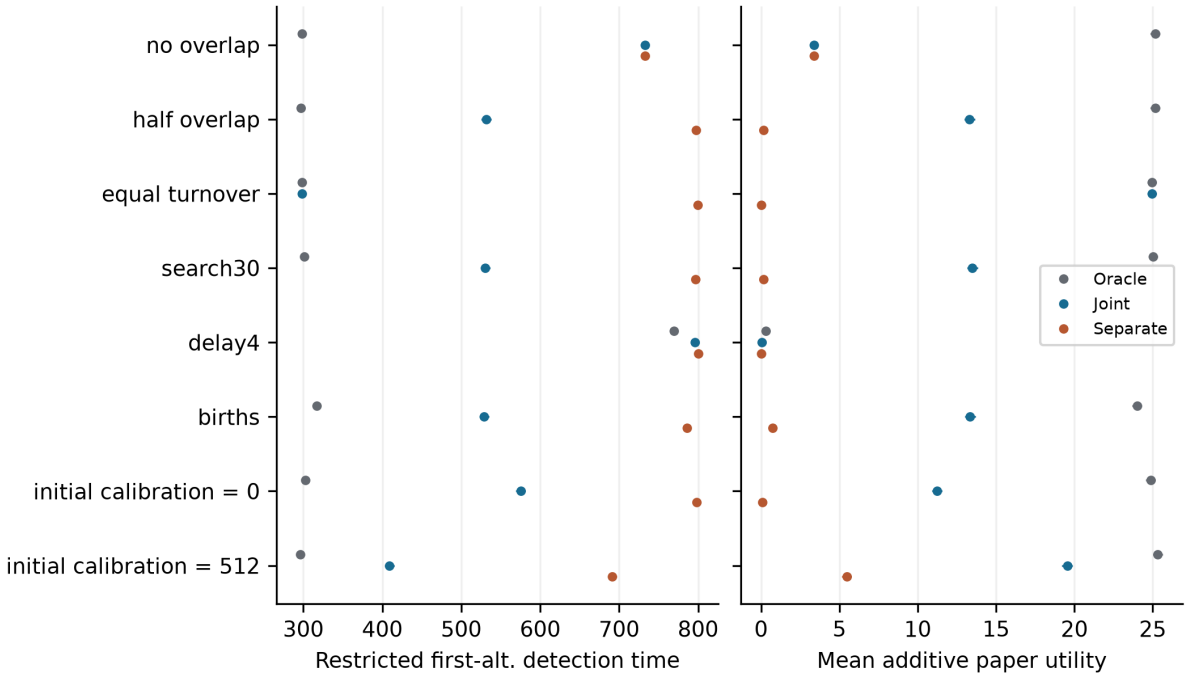


Figure 3: Effect-0.05 diagnostics. Time is first any-alternative discovery from origin, horizon-censored; Figure 1 reports first-stream power. Approximate Monte Carlo intervals are omitted for degenerate samples. Between-case comparisons change several design factors.

power versus 0.000% for local joint EB. With contracting information these become 35.325% and 4.600%. The paired contrasts in Table 2 retain their Monte Carlo uncertainty. In the low-volatility fixed/alternating cell, the coherent/local difference reaches 99.300 percentage points. Thus a local robust calculation can discard much of the available evidence even when its per-label cost geometry is correct. The static, one-sign controls show zero observed coherent/local gain. This is an observed property of these controls, not a general EB equality: the adverse endpoint of a nonmonotone factor can change across labels.

Temporal coherence does not make EB uniformly preferable to other evidence forms. Under high volatility and contracting information, joint linear betting has greater power than coherent EB, for both turnover patterns. Lower volatility and sign changes can reverse that comparison. Likewise, the joint aGRAPA minus joint linear power contrast ranges from -1.225 to 12.900 percentage points across the eight primary cells. Its negative high-volatility/contracting contrasts and positive low-volatility/static contrasts rule out a uniform empirical advantage. Predictable refitting also has a stronger conditional-null requirement than the fixed-grid comparisons.

Local joint versus local separate EB gains range from 0.000 to 98.900 percentage points. These are a different ablation from coherent versus local joint EB. Both must be held separate from changing the underlying betting rule or the cost-calibration schedule. The results identify mechanisms within the declared synthetic family, rather than a universally best sequential procedure.

Weak effects remain difficult. Across the eight effect-0.02 cells, coherent EB first-stream rejection never exceeds 1.825%. The stronger effect-0.05 gains therefore do not establish sensitive detection of small improvements at this horizon, even inside the synthetic family.

Restricted detection time and subsequent paper utility follow the same implementation's recorded allocation rule, not a funded portfolio. Saturated end-of-horizon power does not imply identical detection times or utility; those endpoints remain in the complete result tables. The prefix-weak diagnostic has no first-stream rejections for any method. Its family rejection counts are one for oracle linear, two for oracle aGRAPA and zero for the other methods, out of 4,000

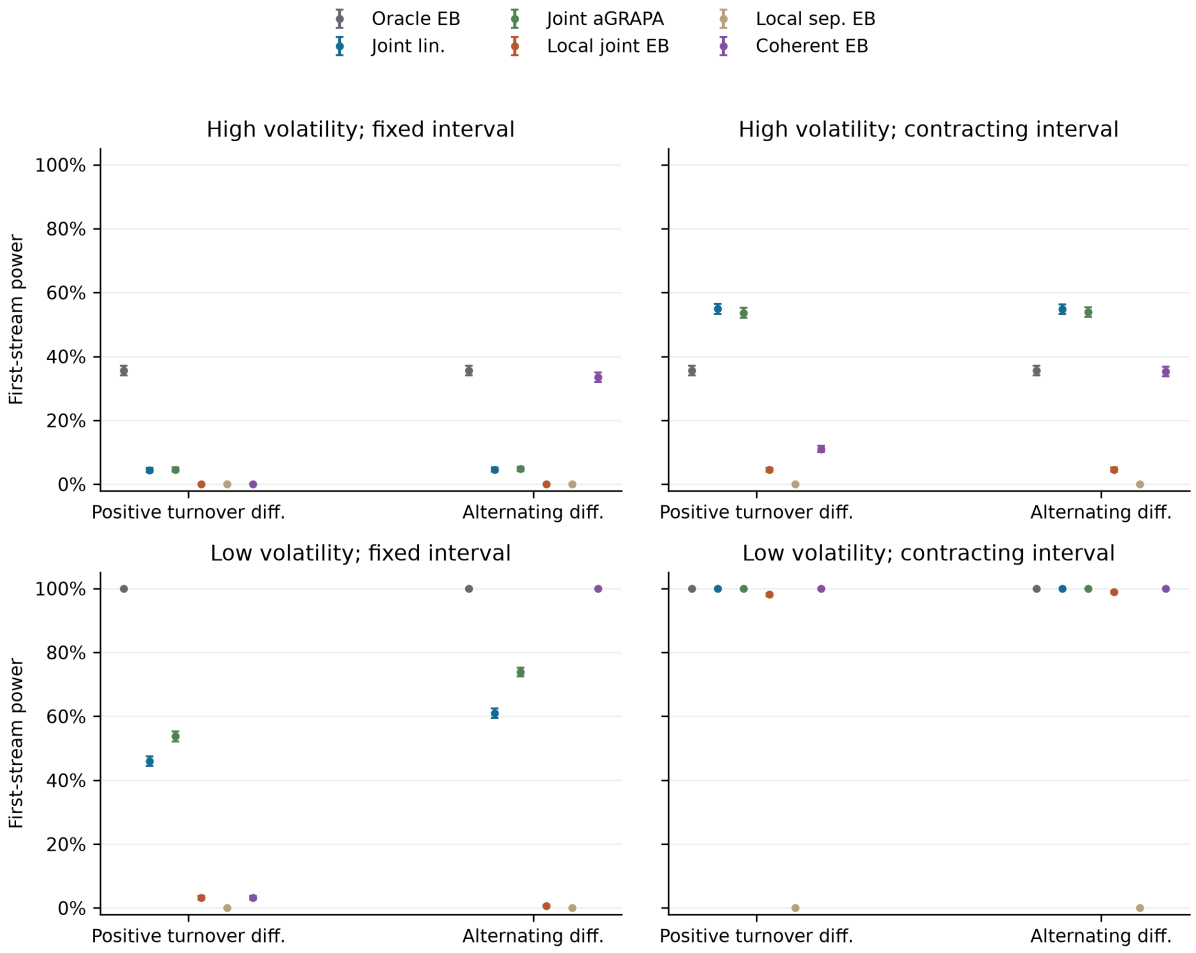


Figure 4: Follow-up effect-0.05 factorial comparison. Points have within-case 95% Wilson intervals over 4,000 worlds. Oracle EB knows the true cost; its predictable centers match the other EB methods. Static and contracting information use different calibration schedules. Factorial cells share seeded innovations and must not be pooled as independent replications.

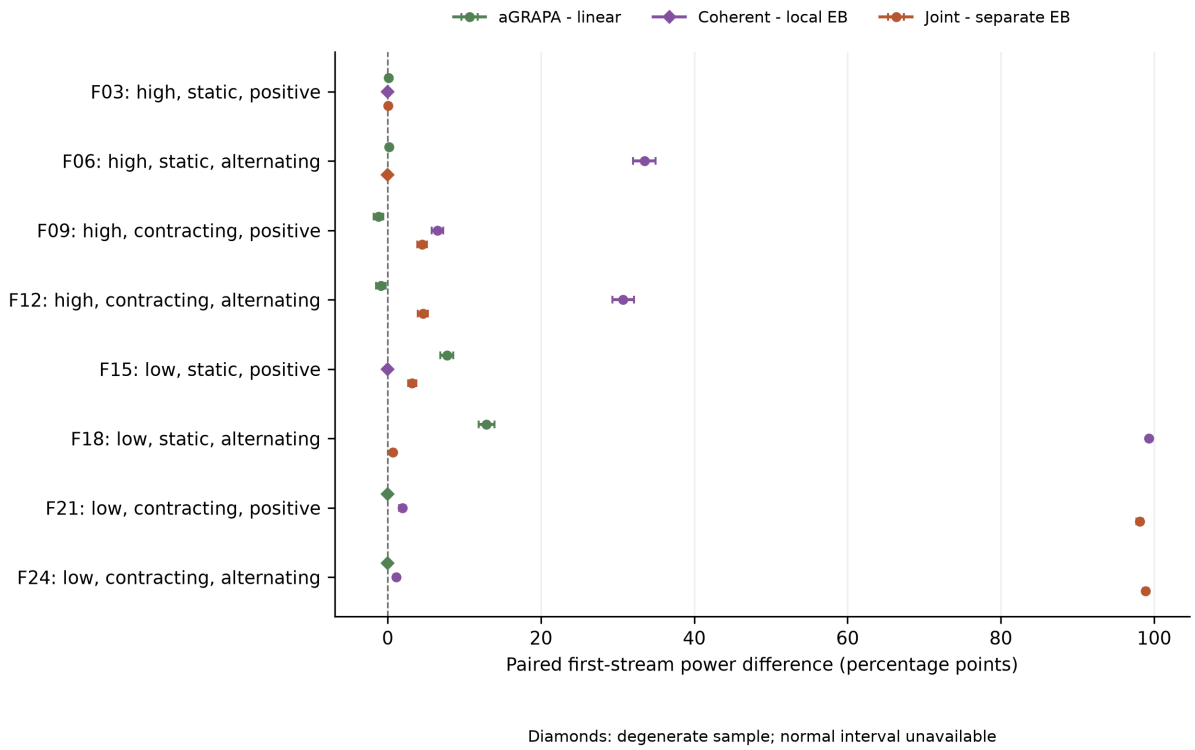


Figure 5: The three primary within-case ablations, with paired approximate 95% normal Monte Carlo intervals. Intervals are unadjusted descriptive summaries; diamonds denote degenerate samples for which that normal interval is unavailable. Every registered contrast, including weaker effects, is retained in the machine-readable supplement.

Case	aGRAPA - linear	Coherent - local EB	Joint - separate EB
F03	0.12 [-0.20, 0.45]	0.00 ‡	0.03 [-0.02, 0.07]
F06	0.18 [-0.15, 0.50]	33.50 [32.04, 34.96]	0.00 ‡
F09	-1.23 [-1.85, -0.60]	6.48 [5.71, 7.24]	4.50 [3.86, 5.14]
F12	-0.88 [-1.50, -0.25]	30.73 [29.30, 32.15]	4.60 [3.95, 5.25]
F15	7.72 [6.90, 8.55]	0.00 ‡	3.20 [2.65, 3.75]
F18	12.90 [11.86, 13.94]	99.30 [99.04, 99.56]	0.70 [0.44, 0.96]
F21	0.00 ‡	1.90 [1.48, 2.32]	98.10 [97.68, 98.52]
F24	0.00 ‡	1.10 [0.78, 1.42]	98.90 [98.58, 99.22]

Table 2: Prespecified effect-0.05 contrasts in first-stream power, in percentage points. Intervals are paired, approximate, unadjusted 95% normal Monte Carlo intervals across worlds. ‡ marks a degenerate sample with an unavailable normal interval, not known zero uncertainty. Case definitions appear in the supplement.

Policy	Return (%)	Max DD (%)	Exposure (%)	Trades	Turnover
Fixed EMA 50/200	2.61	-20.05	26.8	31	13.54
Adaptive	2.61	-20.05	26.8	31	13.54
Ensemble	-2.21	-20.09	25.7	88	14.68
Cash	0.00	0.00	0.0	0	0.00
Buy and hold	-11.27	-53.45	100.0	0	0.00
Volatility target	-6.30	-36.73	51.5	20	2.26
Slow resizing	4.84	-18.50	26.3	27	12.75
Fixed EMA 25/100	-5.17	-22.65	24.4	57	25.51
Fixed EMA 100/400	-4.63	-21.26	26.8	17	6.65

Table 3: Primary retrospective four-hour policies at 25 bps per side. Exposure is average closing weight; turnover sums absolute traded notional divided by pre-trade wealth. Holdings are inherited and terminal holdings marked.

worlds each. These low frequencies do not validate aGRAPA under that weaker null: Section 2.4 gives its scope and a deterministic counterexample to general predictable refitting.

Case kernels total 2,542.6 seconds on the recorded host. Named persistent state uses 24,220,080 bytes per case; this is not peak process memory and runtime excludes serialization. These observations describe reproducibility costs, not controlled speed comparisons. All 250 method summaries, 150 paired contrasts, calibration diagnostics and source-linked array outputs are retained.

4.4 No demonstrated adaptive Bitcoin edge

The primary four-hour period has 3,948 scored candles, no missing scored intervals and no feature pauses. The selector makes no switches. Its full Binance history has 27 decisions, all retaining EMA 50/200. Fixed/adaptive raw paths are exactly identical, with the same 2.607% primary return (Table 3). The ensemble returns -2.211%. These point estimates do not establish adjusted superiority over the controls.

Bitstamp has two pre-window switches and different inherited wealth, but no scored-window switches. Its rebased paths agree to floating-point precision. Differences around 10^{-16} produce some raw cost-grid sign changes; these are not economic break-even costs. A post-outcome note preserves raw artifacts and marks per-bar differences within 10^{-12} as numerical identity. Binance's zero bootstrap interval and Bitstamp's roundoff intervals resample an identical historical difference, not a known zero future advantage.

On the common 243-day window, hourly and daily adaptive annualized mean-log differences are -9.653% and -6.681% at base cost. The hourly approximate 30-day interval is

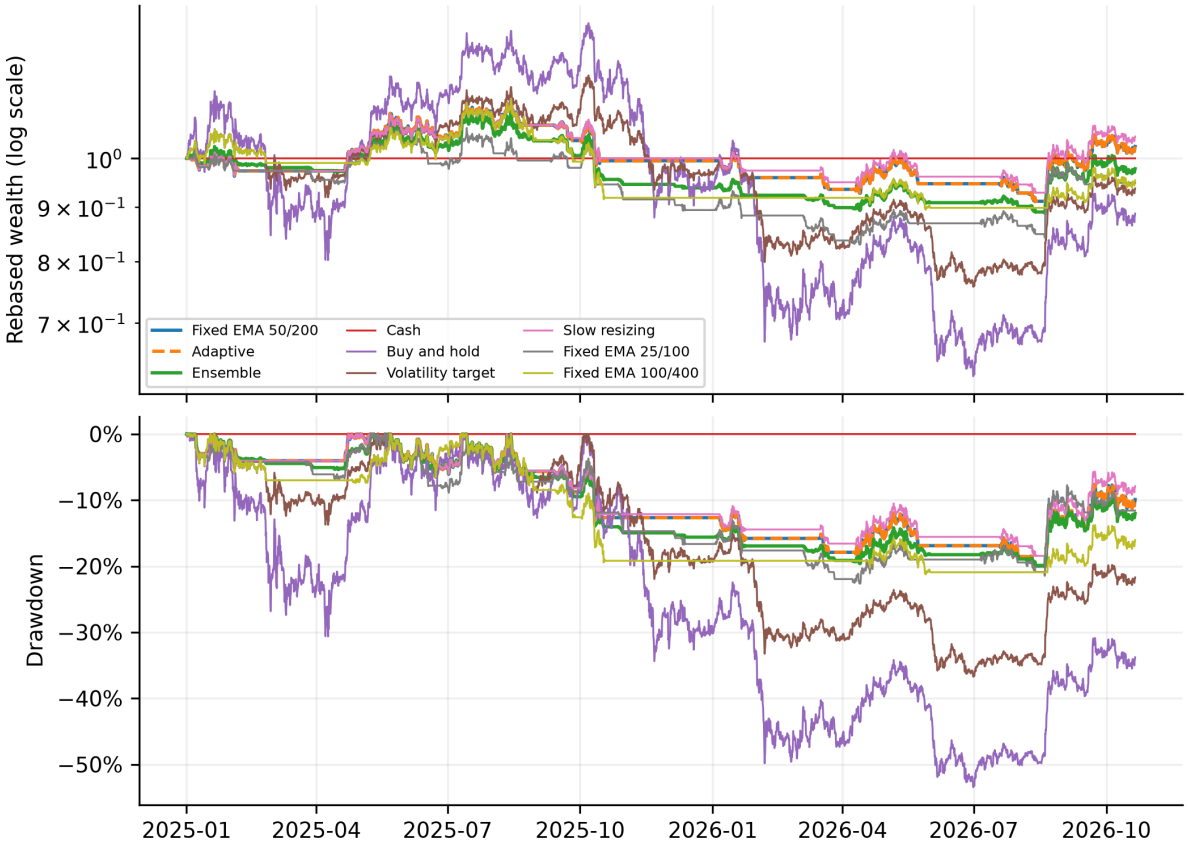


Figure 6: Primary rebased wealth and drawdown at assumed 25 bps. Adaptive and fixed lines coincide. Passive and risk-controlled books have different exposures; no line represents verified historical fills.

Case	Nominal blocks	Observed	Expected resample
Hourly	8.1	-9.653%	-11.150%
Daily	8.1	-6.681%	-7.256%

Table 4: Primary 25 bps matched-window mean log differences, annualized linearly. The expected resample averages exactly over all possible block starts, including the final truncated block. Nominal block count is not an independent sample size.

Case	Annual log diff. (%)	30-day interval (%)	Switches
Binance primary 4h	0.00†	identical path†	0
Binance matched 4h	0.00†	identical path†	0
Binance matched 1h	-9.65	[-32.18, 2.61]	2
Binance matched 1d	-6.68	[-22.42, 5.04]	3
Bitstamp primary 4h	0.00†	identical path†	0
Bitstamp matched 4h	0.00†	identical path†	0

Table 5: Base-cost paired results. † denotes historical numerical identity, not zero future uncertainty. Other intervals are approximate, retrospective and unadjusted.

$[-32.175\%, 2.610\%]$; the daily interval is $[-22.419\%, 5.042\%]$. Both span losses and gains, including the 2% materiality threshold. Neither material gain nor stable loss is established. There are two hourly and three daily switches. These are dependent market observations, not independent replications or evidence of a universally best timeframe.

There are only 8.1 nominal 30-day blocks in this 243-day window. This does not validate stability or finite-sample coverage. A post-outcome deterministic diagnostic in Table 4 shows the finite-sample edge weighting of the specified non-circular resampler: its expected resampled mean differs from the observed mean. These are not new intervals, and no better-looking correction replaces the registered percentile results.

All five costs leave four-hour comparisons identical and hourly/daily point estimates negative. Neither nonidentical case has a cost-grid sign-change bracket. The hourly deficit shrinks slightly at larger costs, illustrating why reference costs must enter the paired contrast. Direct fee-drag differences are small relative to the hourly/daily price/holdings differences; this is accounting description, not causal attribution. All cost/control outcomes are retained in the supplement.

5 Interpretation, limitations and reproducibility

Sequential error control, historical economic performance and execution feasibility answer different questions. Cost calibration assumptions are substantive restrictions, not facts supplied by OHLCV data. The synthetic construction isolates paired-cost uncertainty under known truth; it cannot validate a common stationary market cost parameter, nor show that selected Bitcoin strategies satisfy a bounded conditional null.

Likewise, a confidence interval from one retrospective price history does not create additional independent markets. Native candle frequencies differ in outage history as well as discretization, and a strict reset policy sacrifices history to avoid manufacturing tradable bars. USD and USDT quote risks remain unmodeled. No capacity or actual fill inference is supported by these inputs. Zero interest on cash is a modeling choice that changes relative performance.

The practical result is a negative one for the tested four-hour EMA selector: the primary procedure never departs from its reference, and the nonidentical timeframe checks do not establish material improvement. The synthetic power advantage depends on a defensible shared cost coefficient. It is not evidence that this cost structure has been calibrated for the Bitcoin application. Improving the statistical comparison and improving the trading rule are distinct

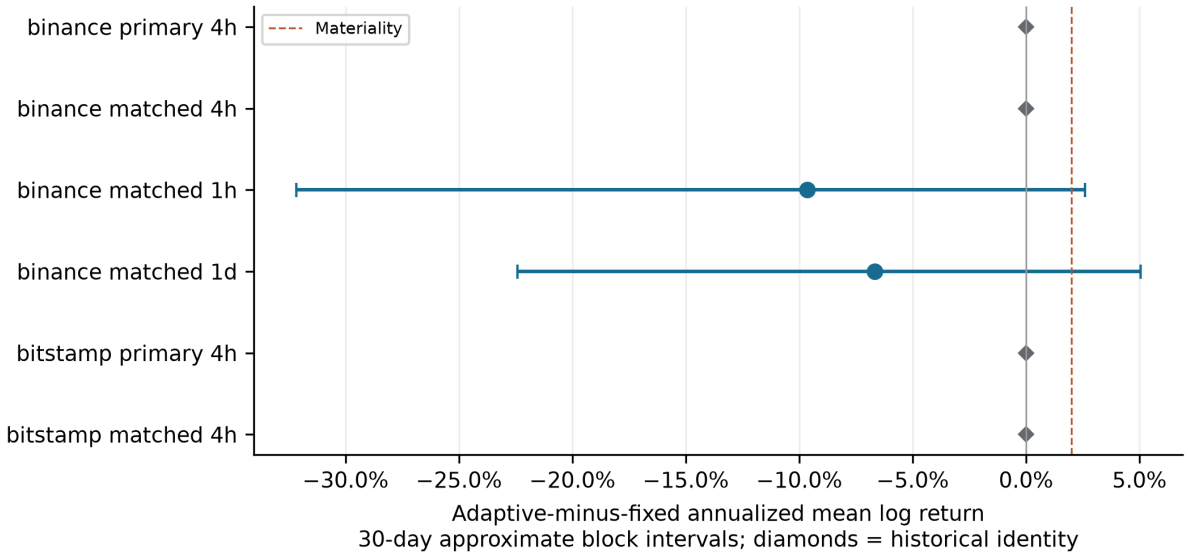


Figure 7: All six prespecified application cases. Diamonds mark historical identity. Intervals are not sequential or research-selection-adjusted evidence. The dashed line is annual log materiality 0.02.

tasks.

The novelty claim is limited to the completed diagnostic findings and their auditable connection to the application. The underlying bounded betting, coverage composition and robust-cost identities are established. The follow-up shows that local cost pessimism, temporal information loss and evidence-form choice have different empirical consequences. Its matched profiling gain is not a new composite-inference theorem. Large synthetic gains and a negative market result can coexist because the market application does not establish the synthetic cost assumptions. Further claims require new prespecified experiments and nearest-work review.

The implementation uses locked Python dependencies, deterministic seeds, synthetic accounting/leakage tests, immutable inputs, source snapshots, configuration/runtime fingerprints and a transactional SQLite attempt ledger. Publication and smoke runs are labeled separately. Freeze creation for the market application checks feature coverage without evaluating portfolio P&L. All attempts, including failures and partial runs, are retained. A local freeze is not an independently witnessed preregistration or a mechanism preventing human or model prior knowledge.

Availability and reproducibility. The accompanying research distribution preserves the exact source ZIPs used by both synthetic cohorts and the market study, specifications, protocols, hashes, all case outputs and full attempt-ledger snapshots. Generated displays contain 216 baseline and 250 follow-up method/case rows, 150 follow-up paired contrasts, 270 market policy/cost rows and 30 primary paired cost rows. AI-assisted internal consistency checks recomputed world summaries, accounting, timing, selector behavior and paired intervals. Check scopes and source fingerprints are retained. These automated checks are not independent human peer review or a human-author approval record. The declarations identify the tool, its substantive roles and the limits of the recorded model information. The documented single entry point verifies preserved evidence, rebuilds this paper, or replays all studies from their archived sources. The completed recorded-host replay reproduced the saved scientific arrays and numerical reports exactly. Its combined invocation reverified an earlier full baseline replay and freshly recomputed the follow-up; this creates no new independent worlds or market observations. Final-paper sources are bound to a separate completed rebuild, with rendered-page QA and local archive checks recorded alongside the replication instructions. The submission companion separates

a LaTeX source bundle from a code/results replication bundle. Raw vendor candles are not included in the public replication bundle. An explicit results-only mode rebuilds displays and the manuscript without claiming a raw-data audit or a fresh computational replay. Exact-input restoration verifies the original hashes and reconciles normalized candles against raw records before a full replay; revised downloads fail that check rather than being relabeled as the original snapshot. Bibliographic access limits and historical revision risk remain explicit. If revisions prevent recovery, the corresponding author can be contacted about permitted access to preserved snapshots. The provider-specific reuse exceptions are stated below.

6 Conclusion

Shared mean-cost structure can preserve substantial sequential detection power in bounded paired experiments, while a superficially conservative upper-cost shortcut can give false evidence when the reference trades more. Temporal profiling can recover evidence discarded by local bounds, but method rankings depend on volatility, calibration and turnover patterns. These are conditional established-method findings, with calibration and information requirements made explicit. The audited Bitcoin application demonstrates no incremental four-hour adaptive performance and no sufficiently precise material gain in its timeframe checks. The result supports disciplined validation of the bot, not a claim that this implementation has found a profitable adaptive edge.

Declarations

Author. Abdullah Hamza directed the research and is the corresponding author. Responsibility for the submitted work rests with the named author.

AI assistance and responsibility. OpenAI Codex was used for literature research, experimental design, code preparation and testing, statistical analysis, interpretation, figure generation, manuscript drafting and revision, and reproducibility and release checks. Exact model versions were not consistently recorded. The human author retains responsibility for the final work. Automated checks and AI-assisted reviews are not independent human peer review or evidence of completed human-author review.

Funding and competing interests. The author reports no external funding and no competing interests. The study uses synthetic experiments and public exchange records, with no human-subject or private account data.

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References

[1] Yo Joong Choe and Aaditya Ramdas (2024). Comparing Sequential Forecasters. *Operations Research*, 72(4):1368–1387. doi:10.1287/opre.2021.0792.

[2] Sebastian Arnold, Georgios Gavrilopoulos, Benedikt Schulz and Johanna Ziegel (2026). Sequential model confidence sets. *Journal of the Royal Statistical Society Series B: Statistical Methodology*. doi:10.1093/jrsssb/qkag066.

[3] Ian Waudby-Smith and Aaditya Ramdas (2024). Estimating means of bounded random variables by betting. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 86(1):1–27. doi:10.1093/jrsssb/qkad009.

[4] Steven R. Howard, Aaditya Ramdas, Jon McAuliffe and Jasjeet Sekhon (2021). Time-uniform, nonparametric, nonasymptotic confidence sequences. *The Annals of Statistics*, 49(2):1055–1080. doi:10.1214/20-AOS1991.

[5] Ziyu Xu and Aaditya Ramdas (2024). Online multiple testing with e-values. *Proceedings of the 27th International Conference on Artificial Intelligence and Statistics, PMLR*, 238:3997–4005. Source and version.

[6] Jinjin Tian and Aaditya Ramdas (2021). Online control of the familywise error rate. *Statistical Methods in Medical Research*, 30(4):976–993. doi:10.1177/0962280220983381.

[7] Zhanyi Jiao, Qiuqi Wang and Yimiao Zhao (2026). Comparative e-backtests for general risk measures. *arXiv preprint*, arXiv:2511.05840v2. Source and version.

[8] Xiaoquan Liu (2026). Anytime-Valid Tests of Factor Alphas. *SSRN working paper*. doi:10.2139/ssrn.7357270. Working paper; only indexed abstract inspected.

[9] Rany Stephan (2026). Sequential Tradeability Testing for Alpha Signals. *SSRN working paper*. doi:10.2139/ssrn.6922558. Working paper; only indexed abstract inspected.

[10] Larry Wasserman, Aaditya Ramdas and Sivaraman Balakrishnan (2020). Universal inference. *Proceedings of the National Academy of Sciences*, 117(29):16880–16890. doi:10.1073/pnas.1922664117.

[11] Aaditya Ramdas, Johannes Ruf, Martin Larsson and Wouter M. Koolen (2022). Testing exchangeability: Fork-convexity, supermartingales and e-processes. *International Journal of Approximate Reasoning*, 141:83–109. doi:10.1016/j.ijar.2021.06.017.

[12] A. Ben-Tal and A. Nemirovski (1998). Robust Convex Optimization. *Mathematics of Operations Research*, 23(4):769–805. doi:10.1287/moor.23.4.769.

[13] Trinh Le and Ummul Ruthbah (2023). Trend-following Strategies for Crypto Investors. *Monash Centre for Financial Studies author manuscript*. Source and version.

[14] Andrei Bysik and Robert Ślepaczuk (2026). Machine Learning-Based Bitcoin Trading Under Transaction Costs: Evidence From Walk-Forward Forecasting. *arXiv preprint*, arXiv:2606.00060v1. Source and version.

[15] Yo Joong Choe and Aaditya Ramdas (2026). Combining Evidence Across Filtrations. *arXiv preprint*, arXiv:2402.09698v5. Source and version.

[16] Hans R. Künsch (1989). The Jackknife and the Bootstrap for General Stationary Observations. *The Annals of Statistics*, 17(3):1217–1241. doi:10.1214/aos/1176347265.

[17] Jing Li and Haibin Zhu (2026). Sequential Confidence Sets for Coverage-Constrained Conformal Model Selection. *arXiv preprint*, arXiv:2609.28522v1. Source and version.

[18] Halbert White (2000). A Reality Check for Data Snooping. *Econometrica*, 68(5):1097–1126. doi:10.1111/1468-0262.00152.

[19] Peter R. Hansen, Asger Lunde and James M. Nason (2011). The Model Confidence Set. *Econometrica*, 79(2):453–497. doi:10.3982/ECTA5771.

A Accounting identities

Let cash C, coin holdings Q, execution price P, pre-trade wealth $V = C + QP$, desired post-fee risky fraction w, and one-sided cost fraction f be given. Signed traded notional z solves

$$QP + z = w(V - f|z|), \qquad z = \frac{wV - QP}{1 + \operatorname{sign}(wV - QP) f w}.$$

Cash becomes $C - z - f|z|$ and coins become $Q + z/P$. The buy and sell denominators differ. With $0 \le w \le 1$ and $0 \le f < 1$, the target is unlevered and solvent. Exact full exits and full investments set the relevant residual coin/cash balance to zero to avoid floating-point phantom transitions.

For close-to-close interval i, define V_i^- from previous cash/coins marked at the current open, V_i^+ from current post-trade cash/coins marked there, and V_i at the close. Then

$$\log \frac{V_i}{V_{i-1}} = \log \frac{V_i^-}{V_{i-1}} + \log \frac{V_i^+}{V_i^-} + \log \frac{V_i}{V_i^+}, \qquad V_i^- - V_i^+ = f|z_i|.$$

The first term includes an unobserved outage interval if observations are missing; the identity does not infer the path within it. Score-window returns include the position inherited at the preceding close, not a flat-book restart.

B Exact invalidity examples

These examples diagnose assumptions rather than estimate market frequencies. Their exact calculations are saved separately from Monte Carlo experiments.

A smaller history can hide relevant cost information. Draw one persistent latent $Y = 1/19$ with probability 0.95 and $Y = -1$ with probability 0.05. Let every observable lower proxy equal $1/19$, while each latent utility equals Y. The observable history never reveals Y, and the latent unconditional/observable-history mean is zero. A simultaneous domination event has probability 0.95, yet the positive proxy crosses any fixed betting threshold with probability one. After one matured latent label, the richer oracle history knows Y, and the conditional null fails on the positive branch. This is why high-probability coverage and a null in an insufficient history do not establish Proposition 1.

Pointwise coverage is insufficient. Let $D_i = 0$. Independently at each of 800 steps, a purported lower proxy is one with probability 0.01 and zero otherwise. Every pointwise domination claim has 99% coverage, but simultaneous failure probability is $1 - 0.99^{800}$. With fixed $\lambda = 0.8$, six failures suffice to cross 20, so false rejection is at least $\mathbb{P}\{\text{Binomial}(800, 0.01) \ge 6\}$. A one-percent global error budget cannot be assigned to that sequence of pointwise intervals.

Reusing a screened label creates evidence from selection. The maximum of ten independent fair signs has mean $1 - 2^{1-10}$, although every unscreened sign has mean zero. Repeatedly selecting that observed maximum and pretending it is a new null trial yields an expected linear factor $1 + 0.8(1 - 2^{1-10}) > 1$. Six all-positive selected batches cross 20 with probability $(1 - 2^{-10})^6$. Candidate birth must not recycle prescreen outcomes.

Upper costs on both sides can be optimistic. If $q^A = 0$, $q^R = 1$, $G = -\theta$, and $m = 0$, then the true difference is zero. The symmetric-upper expression is $u - \theta > 0$ whenever $u > \theta$. The correct paired lower expression uses ℓ, giving $\ell - \theta \le 0$ on coverage. Reference costs belong to the same contrast as strategy costs.

Labels must not choose their own bets. For a fair increment $D \in \{-1/2, 1/2\}$, a bet $\lambda = 1/2$ chosen only when the current increment is already known positive has expected factor 1.125, not one. Related delayed-label constructions can hide the same anticipation error. Issued bets and intervals are frozen before labels mature in the implementation.

C Temporal coherence: a reviewed scope boundary

For one fixed interval $[\ell, u]$, independently minimizing each period's affine utility can be more pessimistic than retaining the same nuisance coefficient throughout history. In cumulative linear utility the excess cost is

$$\frac{u-\ell}{2}\left(\sum_i |d_i| - \left|\sum_i d_i\right|\right) = (u-\ell)\min\left(\sum_i d_i^+, \sum_i d_i^-\right).$$

The gain vanishes for one-sign turnover contrasts at a static interval. Whole-path profiling and comparison at the true nuisance parameter are established principles [10, 11]. The separately specified follow-up in Sections 2.4 and 3.2 distinguishes sign changes, interval contraction and the underlying evidence form. It does not replace the frozen v1 experiment.

D All registered synthetic results

Table 6: All synthetic outcomes. Rates and Wilson intervals are percentages. Time is restricted first-alternative detection time (calendar steps, horizon-censored); utility is mean materiality-adjusted paper utility.

Case	Method	Null FWER [95% CI]	First power [95% CI]	Time	Utility
C01	Oracle	0.700 [0.485, 1.010]	–	–	0.006
C01	Joint	0.000 [0.000, 0.096]	–	–	0.000
C01	Separate	0.000 [0.000, 0.096]	–	–	0.000
C01	Plugin	1.175 [0.885, 1.559]	–	–	-0.002
C01	Upper both	0.000 [0.000, 0.096]	–	–	0.000
C01	Joint H	0.000 [0.000, 0.096]	–	–	0.000
C01	Separate H	0.000 [0.000, 0.096]	–	–	0.000
C01	Fixed time	0.000 [0.000, 0.096]	–	–	0.000
C01	Peeked	0.000 [0.000, 0.096]	–	–	0.000
C02	Oracle	1.000 [0.735, 1.359]	–	–	0.003
C02	Joint	0.025 [0.004, 0.141]	–	–	0.002
C02	Separate	0.000 [0.000, 0.096]	–	–	0.000
C02	Plugin	1.050 [0.778, 1.416]	–	–	0.004
C02	Upper both	0.025 [0.004, 0.141]	–	–	0.002
C02	Joint H	0.000 [0.000, 0.096]	–	–	0.000
C02	Separate H	0.000 [0.000, 0.096]	–	–	0.000
C02	Fixed time	0.000 [0.000, 0.096]	–	–	0.000
C02	Peeked	0.000 [0.000, 0.096]	–	–	0.000
C03	Oracle	0.675 [0.464, 0.980]	–	–	0.009
C03	Joint	0.675 [0.464, 0.980]	–	–	0.009
C03	Separate	0.000 [0.000, 0.096]	–	–	0.000
C03	Plugin	0.675 [0.464, 0.980]	–	–	0.009
C03	Upper both	0.675 [0.464, 0.980]	–	–	0.009
C03	Joint H	0.000 [0.000, 0.096]	–	–	0.000
C03	Separate H	0.000 [0.000, 0.096]	–	–	0.000
C03	Fixed time	0.000 [0.000, 0.096]	–	–	0.000
C03	Peeked	0.000 [0.000, 0.096]	–	–	0.000
C04	Oracle	0.450 [0.285, 0.710]	7.200 [6.439, 8.043]	736.9	1.260
C04	Joint	0.000 [0.000, 0.096]	0.025 [0.004, 0.141]	799.7	0.009
C04	Separate	0.000 [0.000, 0.096]	0.025 [0.004, 0.141]	799.7	0.009
C04	Plugin	0.650 [0.444, 0.951]	8.850 [8.009, 9.770]	726.8	1.468
C04	Upper both	0.000 [0.000, 0.096]	0.025 [0.004, 0.141]	799.7	0.009
C04	Joint H	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C04	Separate H	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C04	Fixed time	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C04	Peeked	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C05	Oracle	0.400 [0.246, 0.649]	6.675 [5.942, 7.491]	742.6	1.167
C05	Joint	0.025 [0.004, 0.141]	0.450 [0.285, 0.710]	796.2	0.078
C05	Separate	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C05	Plugin	0.425 [0.266, 0.680]	7.175 [6.416, 8.017]	740.1	1.185
C05	Upper both	0.025 [0.004, 0.141]	0.450 [0.285, 0.710]	796.2	0.078
C05	Joint H	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C05	Separate H	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C05	Fixed time	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C05	Peeked	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C06	Oracle	0.450 [0.285, 0.710]	7.325 [6.558, 8.174]	739.3	1.188
C06	Joint	0.450 [0.285, 0.710]	7.325 [6.558, 8.174]	739.3	1.188
C06	Separate	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C06	Plugin	0.450 [0.285, 0.710]	7.325 [6.558, 8.174]	739.3	1.188
C06	Upper both	0.450 [0.285, 0.710]	7.325 [6.558, 8.174]	739.3	1.188
C06	Joint H	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C06	Separate H	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C06	Fixed time	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C06	Peeked	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C07	Oracle	0.350 [0.209, 0.587]	91.900 [91.014, 92.706]	298.4	25.157
C07	Joint	0.000 [0.000, 0.096]	13.425 [12.404, 14.517]	732.4	3.394
C07	Separate	0.000 [0.000, 0.096]	13.425 [12.404, 14.517]	732.4	3.394
C07	Plugin	0.725 [0.505, 1.039]	90.650 [89.708, 91.514]	301.9	24.948
C07	Upper both	0.000 [0.000, 0.096]	13.425 [12.404, 14.517]	732.4	3.394
C07	Joint H	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C07	Separate H	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C07	Fixed time	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C07	Peeked	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C08	Oracle	0.325 [0.190, 0.555]	92.775 [91.931, 93.537]	297.1	25.159

Case	Method	Null FWER [95% CI]	First power [95% CI]	Time	Utility
C08	Joint	0.000 [0.000, 0.096]	55.950 [54.407, 57.482]	531.9	13.306
C08	Separate	0.000 [0.000, 0.096]	0.525 [0.344, 0.801]	797.2	0.143
C08	Plugin	0.350 [0.209, 0.587]	92.375 [91.511, 93.157]	298.0	25.094
C08	Upper both	0.000 [0.000, 0.096]	55.950 [54.407, 57.482]	531.9	13.306
C08	Joint H	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C08	Separate H	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C08	Fixed time	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C08	Peeked	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C09	Oracle	0.350 [0.209, 0.587]	92.425 [91.564, 93.205]	298.3	24.951
C09	Joint	0.350 [0.209, 0.587]	92.425 [91.564, 93.205]	298.3	24.951
C09	Separate	0.000 [0.000, 0.096]	0.025 [0.004, 0.141]	799.9	0.005
C09	Plugin	0.350 [0.209, 0.587]	92.425 [91.564, 93.205]	298.3	24.951
C09	Upper both	0.350 [0.209, 0.587]	92.425 [91.564, 93.205]	298.3	24.951
C09	Joint H	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C09	Separate H	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C09	Fixed time	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C09	Peeked	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C10	Oracle	0.525 [0.344, 0.801]	0.475 [0.304, 0.741]	793.9	0.021
C10	Joint	0.525 [0.344, 0.801]	0.475 [0.304, 0.741]	793.9	0.021
C10	Separate	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C10	Plugin	0.525 [0.344, 0.801]	0.475 [0.304, 0.741]	793.9	0.021
C10	Upper both	0.525 [0.344, 0.801]	0.475 [0.304, 0.741]	793.9	0.021
C10	Joint H	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C10	Separate H	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C10	Fixed time	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C10	Peeked	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C11	Oracle	0.575 [0.383, 0.861]	–	–	0.002
C11	Joint	0.025 [0.004, 0.141]	–	–	0.001
C11	Separate	0.000 [0.000, 0.096]	–	–	0.000
C11	Plugin	0.750 [0.526, 1.069]	–	–	-0.002
C11	Upper both	0.025 [0.004, 0.141]	–	–	0.001
C11	Joint H	0.000 [0.000, 0.096]	–	–	0.000
C11	Separate H	0.000 [0.000, 0.096]	–	–	0.000
C11	Fixed time	0.000 [0.000, 0.096]	–	–	0.000
C11	Peeked	0.000 [0.000, 0.096]	–	–	0.000
C12	Oracle	0.325 [0.190, 0.555]	87.225 [86.155, 88.224]	300.9	25.011
C12	Joint	0.000 [0.000, 0.096]	42.300 [40.777, 43.838]	530.6	13.486
C12	Separate	0.000 [0.000, 0.096]	0.300 [0.172, 0.524]	796.8	0.158
C12	Plugin	0.450 [0.285, 0.710]	86.725 [85.638, 87.741]	301.8	24.967
C12	Upper both	0.000 [0.000, 0.096]	42.300 [40.777, 43.838]	530.6	13.486
C12	Joint H	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C12	Separate H	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C12	Fixed time	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C12	Peeked	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C13	Oracle	0.150 [0.069, 0.327]	–	–	0.001
C13	Joint	0.000 [0.000, 0.096]	–	–	0.000
C13	Separate	0.000 [0.000, 0.096]	–	–	0.000
C13	Plugin	0.175 [0.085, 0.361]	–	–	0.002
C13	Upper both	0.000 [0.000, 0.096]	–	–	0.000
C13	Joint H	0.000 [0.000, 0.096]	–	–	0.000
C13	Separate H	0.000 [0.000, 0.096]	–	–	0.000
C13	Fixed time	0.000 [0.000, 0.096]	–	–	0.000
C13	Peeked	0.000 [0.000, 0.096]	–	–	0.000
C14	Oracle	0.025 [0.004, 0.141]	6.675 [5.942, 7.491]	769.2	0.300
C14	Joint	0.000 [0.000, 0.096]	0.725 [0.505, 1.039]	796.2	0.037
C14	Separate	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C14	Plugin	0.025 [0.004, 0.141]	7.000 [6.250, 7.833]	767.6	0.323
C14	Upper both	0.000 [0.000, 0.096]	0.725 [0.505, 1.039]	796.2	0.037
C14	Joint H	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C14	Separate H	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C14	Fixed time	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C14	Peeked	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C15	Oracle	0.950 [0.693, 1.301]	–	–	-0.005
C15	Joint	0.025 [0.004, 0.141]	–	–	0.000
C15	Separate	0.000 [0.000, 0.096]	–	–	0.000
C15	Plugin	0.900 [0.651, 1.243]	–	–	-0.008
C15	Upper both	0.025 [0.004, 0.141]	–	–	0.000
C15	Joint H	0.000 [0.000, 0.096]	–	–	0.000
C15	Separate H	0.000 [0.000, 0.096]	–	–	0.000
C15	Fixed time	0.000 [0.000, 0.096]	–	–	0.000
C15	Peeked	0.000 [0.000, 0.096]	–	–	0.000
C16	Oracle	0.000 [0.000, 0.096]	97.825 [97.325, 98.233]	316.9	23.995

Case	Method	Null FWER [95% CI]	First power [95% CI]	Time	Utility
C16	Joint	0.000 [0.000, 0.096]	75.600 [74.245, 76.906]	529.1	13.340
C16	Separate	0.000 [0.000, 0.096]	3.125 [2.629, 3.711]	785.6	0.722
C16	Plugin	0.000 [0.000, 0.096]	97.725 [97.215, 98.143]	318.4	23.934
C16	Upper both	0.000 [0.000, 0.096]	75.600 [74.245, 76.906]	529.1	13.340
C16	Joint H	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C16	Separate H	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C16	Fixed time	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C16	Peeked	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C17	Oracle	0.725 [0.505, 1.039]	–	–	-0.008
C17	Joint	0.000 [0.000, 0.096]	–	–	0.000
C17	Separate	0.000 [0.000, 0.096]	–	–	0.000
C17	Plugin	0.800 [0.567, 1.127]	–	–	-0.004
C17	Upper both	0.000 [0.000, 0.096]	–	–	0.000
C17	Joint H	0.000 [0.000, 0.096]	–	–	0.000
C17	Separate H	0.000 [0.000, 0.096]	–	–	0.000
C17	Fixed time	0.000 [0.000, 0.096]	–	–	0.000
C17	Peeked	0.000 [0.000, 0.096]	–	–	0.000
C18	Oracle	0.325 [0.190, 0.555]	92.825 [91.983, 93.584]	302.7	24.864
C18	Joint	0.000 [0.000, 0.096]	48.975 [47.428, 50.524]	575.5	11.245
C18	Separate	0.000 [0.000, 0.096]	0.325 [0.190, 0.555]	798.2	0.077
C18	Plugin	0.550 [0.363, 0.831]	92.725 [91.878, 93.490]	304.4	24.792
C18	Upper both	0.000 [0.000, 0.096]	48.975 [47.428, 50.524]	575.5	11.245
C18	Joint H	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C18	Separate H	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C18	Fixed time	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C18	Peeked	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C19	Oracle	0.675 [0.464, 0.980]	–	–	-0.001
C19	Joint	0.125 [0.053, 0.292]	–	–	-0.000
C19	Separate	0.000 [0.000, 0.096]	–	–	0.000
C19	Plugin	0.700 [0.485, 1.010]	–	–	0.003
C19	Upper both	0.125 [0.053, 0.292]	–	–	-0.000
C19	Joint H	0.000 [0.000, 0.096]	–	–	0.000
C19	Separate H	0.000 [0.000, 0.096]	–	–	0.000
C19	Fixed time	0.000 [0.000, 0.096]	–	–	0.000
C19	Peeked	0.000 [0.000, 0.096]	–	–	0.000
C20	Oracle	0.475 [0.304, 0.741]	93.125 [92.299, 93.868]	296.2	25.317
C20	Joint	0.050 [0.014, 0.182]	74.925 [73.558, 76.244]	409.1	19.541
C20	Separate	0.025 [0.004, 0.141]	18.425 [17.254, 19.657]	691.5	5.468
C20	Plugin	0.475 [0.304, 0.741]	93.175 [92.351, 93.916]	296.9	25.307
C20	Upper both	0.050 [0.014, 0.182]	74.925 [73.558, 76.244]	409.1	19.541
C20	Joint H	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C20	Separate H	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C20	Fixed time	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C20	Peeked	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C21	Oracle	0.575 [0.383, 0.861]	–	–	0.013
C21	Joint	0.000 [0.000, 0.096]	–	–	0.000
C21	Separate	0.000 [0.000, 0.096]	–	–	0.000
C21	Plugin	0.575 [0.383, 0.861]	–	–	0.017
C21	Upper both	0.000 [0.000, 0.096]	–	–	0.000
C21	Joint H	0.000 [0.000, 0.096]	–	–	0.000
C21	Separate H	0.000 [0.000, 0.096]	–	–	0.000
C21	Fixed time	0.000 [0.000, 0.096]	–	–	0.000
C21	Peeked	0.000 [0.000, 0.096]	–	–	0.000
C22	Oracle	0.775 [0.547, 1.098]	–	–	-0.005
C22	Joint	0.100 [0.039, 0.257]	–	–	0.001
C22	Separate	0.100 [0.039, 0.257]	–	–	0.001
C22	Plugin	43.550 [42.020, 45.092]	–	–	-0.029
C22	Upper both	0.100 [0.039, 0.257]	–	–	0.001
C22	Joint H	0.000 [0.000, 0.096]	–	–	0.000
C22	Separate H	0.000 [0.000, 0.096]	–	–	0.000
C22	Fixed time	0.000 [0.000, 0.096]	–	–	0.000
C22	Peeked	0.000 [0.000, 0.096]	–	–	0.000
C23	Oracle	0.875 [0.630, 1.214]	–	–	0.001
C23	Joint	0.100 [0.039, 0.257]	–	–	-0.000
C23	Separate	0.000 [0.000, 0.096]	–	–	0.000
C23	Plugin	0.975 [0.714, 1.330]	–	–	0.000
C23	Upper both	12.950 [11.945, 14.026]	–	–	-0.040
C23	Joint H	0.000 [0.000, 0.096]	–	–	0.000
C23	Separate H	0.000 [0.000, 0.096]	–	–	0.000
C23	Fixed time	0.000 [0.000, 0.096]	–	–	0.000
C23	Peeked	0.000 [0.000, 0.096]	–	–	0.000
C24	Oracle	0.400 [0.246, 0.649]	92.950 [92.115, 93.703]	298.4	25.144

Case	Method	Null FWER [95% CI]	First power [95% CI]	Time	Utility
C24	Joint	0.000 [0.000, 0.096]	56.100 [54.557, 57.631]	532.3	13.456
C24	Separate	0.000 [0.000, 0.096]	0.700 [0.485, 1.010]	795.0	0.256
C24	Plugin	0.575 [0.383, 0.861]	92.925 [92.089, 93.679]	298.3	25.127
C24	Upper both	9.025 [8.176, 9.953]	99.800 [99.606, 99.899]	171.4	31.406
C24	Joint H	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C24	Separate H	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C24	Fixed time	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000
C24	Peeked	0.000 [0.000, 0.096]	0.000 [0.000, 0.096]	800.0	0.000

Null rates remain null rates in mixed families. No independence among candidates is assumed. Plugin, Upper both and Peeked are unguaranteed comparators. All methods in C22 use an explicitly violated cost model; no proxy guarantee is asserted there. Full utility/time uncertainty and runtime/state-array dimensions are in the accompanying CSV.

Case	Registered specification key
C01	`null_no_overlap`
C02	`null_half_overlap`
C03	`null_equal_turnover`
C04	`effect02_no_overlap`
C05	`effect02_half_overlap`
C06	`effect02_equal_turnover`
C07	`effect05_no_overlap`
C08	`effect05_half_overlap`
C09	`effect05_equal_turnover`
C10	`weak005_equal_turnover`
C11	`null_search30`
C12	`effect05_search30`
C13	`null_delay4`
C14	`effect05_delay4`
C15	`null_births`
C16	`effect05_births`
C17	`null_no_calibration`
C18	`effect05_no_calibration`
C19	`null_calibration512`
C20	`effect05_calibration512`
C21	`weak_prefix_null`
C22	`violated_cost_stationarity`
C23	`reference_more_turnover`
C24	`effect05_reference_more_turnover`

E Complete follow-up results

Case	Specification key
F01	`vhigh_static_initial_positive_e0`
F02	`vhigh_static_initial_positive_e2`
F03	`vhigh_static_initial_positive_e5`
F04	`vhigh_static_initial_alternating_e0`
F05	`vhigh_static_initial_alternating_e2`
F06	`vhigh_static_initial_alternating_e5`
F07	`vhigh_contracting_positive_e0`
F08	`vhigh_contracting_positive_e2`
F09	`vhigh_contracting_positive_e5`
F10	`vhigh_contracting_alternating_e0`
F11	`vhigh_contracting_alternating_e2`
F12	`vhigh_contracting_alternating_e5`
F13	`vlow_static_initial_positive_e0`
F14	`vlow_static_initial_positive_e2`
F15	`vlow_static_initial_positive_e5`
F16	`vlow_static_initial_alternating_e0`
F17	`vlow_static_initial_alternating_e2`
F18	`vlow_static_initial_alternating_e5`
F19	`vlow_contracting_positive_e0`
F20	`vlow_contracting_positive_e2`
F21	`vlow_contracting_positive_e5`
F22	`vlow_contracting_alternating_e0`
F23	`vlow_contracting_alternating_e2`
F24	`vlow_contracting_alternating_e5`
F25	`prefix_weak_diagnostic`

The aGRAPA rows in F25 are an assumption diagnostic without a prefix-weak null guarantee. Other rows retain the scope in Section 2.4; empirical null frequencies alone do not extend it.

Table 9: All separately specified v2 outcomes. Rates and Wilson intervals are percentages. Time is horizon-censored first-stream time; utility is additive materiality-adjusted paper utility.

Case	Method	Null rate [95% CI]	First rej. [95% CI]	Time	Utility
F01	Oracle lin.	0.85 [0.61, 1.19]	0.03 [0.00, 0.14]	799.9	0.000
F01	Joint lin.	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F01	Separate lin.	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F01	Oracle aGRAPA	1.15 [0.86, 1.53]	0.10 [0.04, 0.26]	799.3	-0.012
F01	Joint aGRAPA	0.05 [0.01, 0.18]	0.00 [0.00, 0.10]	800.0	-0.003
F01	Separate aGRAPA	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F01	Oracle EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F01	Local joint EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F01	Local sep. EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F01	Coherent EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F02	Oracle lin.	0.83 [0.59, 1.16]	7.32 [6.56, 8.17]	778.4	0.420
F02	Joint lin.	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F02	Separate lin.	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F02	Oracle aGRAPA	1.12 [0.84, 1.50]	7.30 [6.53, 8.15]	775.0	0.462
F02	Joint aGRAPA	0.05 [0.01, 0.18]	0.00 [0.00, 0.10]	800.0	-0.003
F02	Separate aGRAPA	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F02	Oracle EB	0.00 [0.00, 0.10]	0.03 [0.00, 0.14]	800.0	-0.000
F02	Local joint EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F02	Local sep. EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F02	Coherent EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F03	Oracle lin.	0.83 [0.59, 1.16]	93.30 [92.48, 94.03]	424.1	18.800
F03	Joint lin.	0.00 [0.00, 0.10]	4.42 [3.83, 5.11]	787.3	0.631
F03	Separate lin.	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F03	Oracle aGRAPA	1.12 [0.84, 1.50]	92.70 [91.85, 93.47]	415.8	19.213
F03	Joint aGRAPA	0.05 [0.01, 0.18]	4.55 [3.95, 5.24]	784.2	0.782

Case	Method	Null rate [95% CI]	First rej. [95% CI]	Time	Utility
F03	Separate aGRAPA	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F03	Oracle EB	0.00 [0.00, 0.10]	35.52 [34.06, 37.02]	746.5	2.659
F03	Local joint EB	0.00 [0.00, 0.10]	0.03 [0.00, 0.14]	800.0	0.001
F03	Local sep. EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F03	Coherent EB	0.00 [0.00, 0.10]	0.03 [0.00, 0.14]	800.0	0.001
F04	Oracle lin.	0.85 [0.61, 1.19]	0.03 [0.00, 0.14]	799.9	0.000
F04	Joint lin.	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F04	Separate lin.	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F04	Oracle aGRAPA	1.15 [0.86, 1.53]	0.10 [0.04, 0.26]	799.3	-0.012
F04	Joint aGRAPA	0.03 [0.00, 0.14]	0.00 [0.00, 0.10]	800.0	-0.002
F04	Separate aGRAPA	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F04	Oracle EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F04	Local joint EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F04	Local sep. EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F04	Coherent EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F05	Oracle lin.	0.83 [0.59, 1.16]	7.32 [6.56, 8.17]	778.4	0.420
F05	Joint lin.	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F05	Separate lin.	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F05	Oracle aGRAPA	1.12 [0.84, 1.50]	7.30 [6.53, 8.15]	775.0	0.462
F05	Joint aGRAPA	0.03 [0.00, 0.14]	0.00 [0.00, 0.10]	800.0	-0.002
F05	Separate aGRAPA	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F05	Oracle EB	0.00 [0.00, 0.10]	0.03 [0.00, 0.14]	800.0	-0.000
F05	Local joint EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F05	Local sep. EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F05	Coherent EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F06	Oracle lin.	0.83 [0.59, 1.16]	93.30 [92.48, 94.03]	424.1	18.800
F06	Joint lin.	0.00 [0.00, 0.10]	4.60 [3.99, 5.29]	786.1	0.695
F06	Separate lin.	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F06	Oracle aGRAPA	1.12 [0.84, 1.50]	92.70 [91.85, 93.47]	415.8	19.213
F06	Joint aGRAPA	0.03 [0.00, 0.14]	4.78 [4.16, 5.48]	783.4	0.824
F06	Separate aGRAPA	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F06	Oracle EB	0.00 [0.00, 0.10]	35.58 [34.11, 37.07]	746.5	2.665
F06	Local joint EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F06	Local sep. EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F06	Coherent EB	0.00 [0.00, 0.10]	33.50 [32.05, 34.98]	752.9	2.360
F07	Oracle lin.	0.85 [0.61, 1.19]	0.03 [0.00, 0.14]	799.9	0.000
F07	Joint lin.	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F07	Separate lin.	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F07	Oracle aGRAPA	1.15 [0.86, 1.53]	0.10 [0.04, 0.26]	799.3	-0.012
F07	Joint aGRAPA	0.05 [0.01, 0.18]	0.00 [0.00, 0.10]	800.0	-0.002
F07	Separate aGRAPA	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F07	Oracle EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F07	Local joint EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F07	Local sep. EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F07	Coherent EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F08	Oracle lin.	0.83 [0.59, 1.16]	7.32 [6.56, 8.17]	778.4	0.420
F08	Joint lin.	0.00 [0.00, 0.10]	0.22 [0.12, 0.43]	799.4	0.010
F08	Separate lin.	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F08	Oracle aGRAPA	1.12 [0.84, 1.50]	7.30 [6.53, 8.15]	775.0	0.462
F08	Joint aGRAPA	0.05 [0.01, 0.18]	0.27 [0.15, 0.49]	799.3	0.011
F08	Separate aGRAPA	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F08	Oracle EB	0.00 [0.00, 0.10]	0.03 [0.00, 0.14]	800.0	-0.000
F08	Local joint EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F08	Local sep. EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F08	Coherent EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F09	Oracle lin.	0.83 [0.59, 1.16]	93.30 [92.48, 94.03]	424.1	18.800
F09	Joint lin.	0.00 [0.00, 0.10]	54.87 [53.33, 56.41]	659.1	7.025
F09	Separate lin.	0.00 [0.00, 0.10]	0.62 [0.42, 0.92]	799.2	0.035
F09	Oracle aGRAPA	1.12 [0.84, 1.50]	92.70 [91.85, 93.47]	415.8	19.213
F09	Joint aGRAPA	0.05 [0.01, 0.18]	53.65 [52.10, 55.19]	654.6	7.236
F09	Separate aGRAPA	0.00 [0.00, 0.10]	0.70 [0.48, 1.01]	799.1	0.044
F09	Oracle EB	0.00 [0.00, 0.10]	35.60 [34.13, 37.10]	746.5	2.664
F09	Local joint EB	0.00 [0.00, 0.10]	4.50 [3.90, 5.19]	795.5	0.231
F09	Local sep. EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F09	Coherent EB	0.00 [0.00, 0.10]	10.97 [10.04, 11.98]	787.6	0.629
F10	Oracle lin.	0.85 [0.61, 1.19]	0.03 [0.00, 0.14]	799.9	0.000
F10	Joint lin.	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F10	Separate lin.	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F10	Oracle aGRAPA	1.15 [0.86, 1.53]	0.10 [0.04, 0.26]	799.3	-0.012
F10	Joint aGRAPA	0.05 [0.01, 0.18]	0.00 [0.00, 0.10]	800.0	-0.002
F10	Separate aGRAPA	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F10	Oracle EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000

Case	Method	Null rate [95% CI]	First rej. [95% CI]	Time	Utility
F10	Local joint EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F10	Local sep. EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F10	Coherent EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F11	Oracle lin.	0.83 [0.59, 1.16]	7.32 [6.56, 8.17]	778.4	0.420
F11	Joint lin.	0.00 [0.00, 0.10]	0.27 [0.15, 0.49]	799.5	0.006
F11	Separate lin.	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F11	Oracle aGRAPA	1.12 [0.84, 1.50]	7.30 [6.53, 8.15]	775.0	0.462
F11	Joint aGRAPA	0.05 [0.01, 0.18]	0.27 [0.15, 0.49]	799.2	0.011
F11	Separate aGRAPA	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F11	Oracle EB	0.00 [0.00, 0.10]	0.03 [0.00, 0.14]	800.0	-0.000
F11	Local joint EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F11	Local sep. EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F11	Coherent EB	0.00 [0.00, 0.10]	0.03 [0.00, 0.14]	800.0	-0.000
F12	Oracle lin.	0.83 [0.59, 1.16]	93.30 [92.48, 94.03]	424.1	18.800
F12	Joint lin.	0.00 [0.00, 0.10]	54.77 [53.23, 56.31]	661.5	6.949
F12	Separate lin.	0.00 [0.00, 0.10]	0.60 [0.40, 0.89]	799.4	0.028
F12	Oracle aGRAPA	1.12 [0.84, 1.50]	92.70 [91.85, 93.47]	415.8	19.213
F12	Joint aGRAPA	0.05 [0.01, 0.18]	53.90 [52.35, 55.44]	657.5	7.133
F12	Separate aGRAPA	0.00 [0.00, 0.10]	0.70 [0.48, 1.01]	799.2	0.042
F12	Oracle EB	0.00 [0.00, 0.10]	35.58 [34.11, 37.07]	746.5	2.665
F12	Local joint EB	0.00 [0.00, 0.10]	4.60 [3.99, 5.29]	795.7	0.219
F12	Local sep. EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F12	Coherent EB	0.00 [0.00, 0.10]	35.33 [33.86, 36.82]	747.8	2.594
F13	Oracle lin.	0.05 [0.01, 0.18]	0.00 [0.00, 0.10]	800.0	0.000
F13	Joint lin.	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F13	Separate lin.	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F13	Oracle aGRAPA	0.43 [0.27, 0.68]	0.03 [0.00, 0.14]	800.0	-0.001
F13	Joint aGRAPA	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F13	Separate aGRAPA	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F13	Oracle EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F13	Local joint EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F13	Local sep. EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F13	Coherent EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F14	Oracle lin.	0.05 [0.01, 0.18]	87.85 [86.80, 88.83]	578.1	4.429
F14	Joint lin.	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F14	Separate lin.	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F14	Oracle aGRAPA	0.43 [0.27, 0.68]	94.12 [93.35, 94.81]	490.0	6.200
F14	Joint aGRAPA	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F14	Separate aGRAPA	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F14	Oracle EB	0.00 [0.00, 0.10]	1.85 [1.48, 2.32]	798.8	0.023
F14	Local joint EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F14	Local sep. EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F14	Coherent EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F15	Oracle lin.	0.05 [0.01, 0.18]	100.00 [99.90, 100.00]	201.5	29.961
F15	Joint lin.	0.00 [0.00, 0.10]	45.98 [44.44, 47.52]	685.9	5.703
F15	Separate lin.	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F15	Oracle aGRAPA	0.43 [0.27, 0.68]	100.00 [99.90, 100.00]	161.6	31.968
F15	Joint aGRAPA	0.00 [0.00, 0.10]	53.70 [52.15, 55.24]	631.8	8.387
F15	Separate aGRAPA	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F15	Oracle EB	0.00 [0.00, 0.10]	100.00 [99.90, 100.00]	407.9	19.607
F15	Local joint EB	0.00 [0.00, 0.10]	3.20 [2.70, 3.79]	796.9	0.155
F15	Local sep. EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F15	Coherent EB	0.00 [0.00, 0.10]	3.20 [2.70, 3.79]	796.9	0.155
F16	Oracle lin.	0.05 [0.01, 0.18]	0.00 [0.00, 0.10]	800.0	0.000
F16	Joint lin.	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F16	Separate lin.	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F16	Oracle aGRAPA	0.43 [0.27, 0.68]	0.03 [0.00, 0.14]	800.0	-0.001
F16	Joint aGRAPA	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F16	Separate aGRAPA	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F16	Oracle EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F16	Local joint EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F16	Local sep. EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F16	Coherent EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F17	Oracle lin.	0.05 [0.01, 0.18]	87.85 [86.80, 88.83]	578.1	4.429
F17	Joint lin.	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F17	Separate lin.	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F17	Oracle aGRAPA	0.43 [0.27, 0.68]	94.12 [93.35, 94.81]	490.0	6.200
F17	Joint aGRAPA	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F17	Separate aGRAPA	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F17	Oracle EB	0.00 [0.00, 0.10]	1.85 [1.48, 2.32]	798.8	0.023
F17	Local joint EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F17	Local sep. EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000

Case	Method	Null rate [95% CI]	First rej. [95% CI]	Time	Utility
F17	Coherent EB	0.00 [0.00, 0.10]	1.18 [0.88, 1.56]	799.4	0.011
F18	Oracle lin.	0.05 [0.01, 0.18]	100.00 [99.90, 100.00]	201.5	29.961
F18	Joint lin.	0.00 [0.00, 0.10]	60.98 [59.45, 62.48]	667.1	6.652
F18	Separate lin.	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F18	Oracle aGRAPA	0.43 [0.27, 0.68]	100.00 [99.90, 100.00]	161.6	31.968
F18	Joint aGRAPA	0.00 [0.00, 0.10]	73.88 [72.49, 75.21]	592.6	10.375
F18	Separate aGRAPA	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F18	Oracle EB	0.00 [0.00, 0.10]	100.00 [99.90, 100.00]	407.7	19.618
F18	Local joint EB	0.00 [0.00, 0.10]	0.70 [0.48, 1.01]	799.6	0.018
F18	Local sep. EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F18	Coherent EB	0.00 [0.00, 0.10]	100.00 [99.90, 100.00]	420.9	18.956
F19	Oracle lin.	0.05 [0.01, 0.18]	0.00 [0.00, 0.10]	800.0	0.000
F19	Joint lin.	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F19	Separate lin.	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F19	Oracle aGRAPA	0.43 [0.27, 0.68]	0.03 [0.00, 0.14]	800.0	-0.001
F19	Joint aGRAPA	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F19	Separate aGRAPA	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F19	Oracle EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F19	Local joint EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F19	Local sep. EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F19	Coherent EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F20	Oracle lin.	0.05 [0.01, 0.18]	87.85 [86.80, 88.83]	578.1	4.429
F20	Joint lin.	0.00 [0.00, 0.10]	2.30 [1.88, 2.81]	797.6	0.047
F20	Separate lin.	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F20	Oracle aGRAPA	0.43 [0.27, 0.68]	94.12 [93.35, 94.81]	490.0	6.200
F20	Joint aGRAPA	0.00 [0.00, 0.10]	4.42 [3.83, 5.11]	793.7	0.125
F20	Separate aGRAPA	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F20	Oracle EB	0.00 [0.00, 0.10]	1.85 [1.48, 2.32]	798.8	0.023
F20	Local joint EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F20	Local sep. EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F20	Coherent EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F21	Oracle lin.	0.05 [0.01, 0.18]	100.00 [99.90, 100.00]	201.5	29.961
F21	Joint lin.	0.00 [0.00, 0.10]	100.00 [99.90, 100.00]	328.3	23.598
F21	Separate lin.	0.00 [0.00, 0.10]	12.25 [11.27, 13.30]	790.7	0.460
F21	Oracle aGRAPA	0.43 [0.27, 0.68]	100.00 [99.90, 100.00]	161.6	31.968
F21	Joint aGRAPA	0.00 [0.00, 0.10]	100.00 [99.90, 100.00]	279.3	26.036
F21	Separate aGRAPA	0.00 [0.00, 0.10]	21.22 [19.99, 22.52]	781.0	0.949
F21	Oracle EB	0.00 [0.00, 0.10]	100.00 [99.90, 100.00]	407.8	19.614
F21	Local joint EB	0.00 [0.00, 0.10]	98.10 [97.63, 98.48]	608.3	9.574
F21	Local sep. EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F21	Coherent EB	0.00 [0.00, 0.10]	100.00 [99.90, 100.00]	528.5	13.577
F22	Oracle lin.	0.05 [0.01, 0.18]	0.00 [0.00, 0.10]	800.0	0.000
F22	Joint lin.	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F22	Separate lin.	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F22	Oracle aGRAPA	0.43 [0.27, 0.68]	0.03 [0.00, 0.14]	800.0	-0.001
F22	Joint aGRAPA	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F22	Separate aGRAPA	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F22	Oracle EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F22	Local joint EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F22	Local sep. EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F22	Coherent EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F23	Oracle lin.	0.05 [0.01, 0.18]	87.85 [86.80, 88.83]	578.1	4.429
F23	Joint lin.	0.00 [0.00, 0.10]	1.45 [1.12, 1.87]	798.7	0.026
F23	Separate lin.	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F23	Oracle aGRAPA	0.43 [0.27, 0.68]	94.12 [93.35, 94.81]	490.0	6.200
F23	Joint aGRAPA	0.00 [0.00, 0.10]	3.17 [2.67, 3.76]	796.3	0.076
F23	Separate aGRAPA	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F23	Oracle EB	0.00 [0.00, 0.10]	1.85 [1.48, 2.32]	798.8	0.023
F23	Local joint EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F23	Local sep. EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F23	Coherent EB	0.00 [0.00, 0.10]	1.82 [1.45, 2.29]	798.9	0.021
F24	Oracle lin.	0.05 [0.01, 0.18]	100.00 [99.90, 100.00]	201.5	29.961
F24	Joint lin.	0.00 [0.00, 0.10]	100.00 [99.90, 100.00]	326.8	23.666
F24	Separate lin.	0.00 [0.00, 0.10]	9.90 [9.01, 10.86]	793.5	0.332
F24	Oracle aGRAPA	0.43 [0.27, 0.68]	100.00 [99.90, 100.00]	161.6	31.968
F24	Joint aGRAPA	0.00 [0.00, 0.10]	100.00 [99.90, 100.00]	277.9	26.117
F24	Separate aGRAPA	0.00 [0.00, 0.10]	20.05 [18.84, 21.32]	784.5	0.777
F24	Oracle EB	0.00 [0.00, 0.10]	100.00 [99.90, 100.00]	407.7	19.618
F24	Local joint EB	0.00 [0.00, 0.10]	98.90 [98.53, 99.18]	607.8	9.606
F24	Local sep. EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F24	Coherent EB	0.00 [0.00, 0.10]	100.00 [99.90, 100.00]	410.6	19.470
F25	Oracle lin.	0.03 [0.00, 0.14]	0.00 [0.00, 0.10]	800.0	-0.000

Case	Method	Null rate [95% CI]	First rej. [95% CI]	Time	Utility
F25	Joint lin.	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F25	Separate lin.	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F25	Oracle aGRAPA ⋆	0.05 [0.01, 0.18]	0.00 [0.00, 0.10]	800.0	0.001
F25	Joint aGRAPA ⋆	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F25	Separate aGRAPA ⋆	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F25	Oracle EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F25	Local joint EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F25	Local sep. EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000
F25	Coherent EB	0.00 [0.00, 0.10]	0.00 [0.00, 0.10]	800.0	0.000

⋆ Unsupported under the prefix-weak null: these rates are diagnostics, not a valid 5% FWER test. Each other error guarantee is per comparator, not pooled across methods. Full time/utility uncertainty and all paired contrasts are in the CSV files. The machine-readable companion files `method-results.csv`, `paired-results.csv` and `calibration-compute.csv` retain all endpoints and uncertainty intervals. The display trace binds these files to the 25 completed case reports, their arrays and the exact frozen source.

F Complete market cost sensitivity

Table 10: All market cost sensitivities. Differences are annualized mean log returns in percent; intervals use 30-day blocks.

Case	Cost (bps)	Difference (%)	Interval (%)	Identity
M01	25	0.00	historical identity	yes
M01	0	0.00	historical identity	yes
M01	10	0.00	historical identity	yes
M01	50	0.00	historical identity	yes
M01	100	0.00	historical identity	yes
M02	25	0.00	historical identity	yes
M02	0	0.00	historical identity	yes
M02	10	0.00	historical identity	yes
M02	50	0.00	historical identity	yes
M02	100	0.00	historical identity	yes
M03	25	-9.65	[-32.18, 2.61]	no
M03	0	-9.77	[-32.33, 2.47]	no
M03	10	-9.73	[-32.27, 2.53]	no
M03	50	-9.53	[-32.00, 2.75]	no
M03	100	-9.29	[-31.67, 3.03]	no
M04	25	-6.68	[-22.42, 5.04]	no
M04	0	-6.62	[-21.15, 4.19]	no
M04	10	-6.65	[-21.73, 4.61]	no
M04	50	-6.74	[-24.02, 5.91]	no
M04	100	-6.85	[-27.36, 7.95]	no
M05	25	0.00	historical identity	yes
M05	0	0.00	historical identity	yes
M05	10	0.00	historical identity	yes
M05	50	0.00	historical identity	yes
M05	100	0.00	historical identity	yes
M06	25	0.00	historical identity	yes
M06	0	0.00	historical identity	yes
M06	10	0.00	historical identity	yes
M06	50	0.00	historical identity	yes
M06	100	0.00	historical identity	yes

Numerical identity is defined by maximum absolute per-bar paired log difference at most 10^{-12}. The full CSV preserves raw floating-point values. Identity rows have no economically interpretable break-even sign bracket.

Case	Market specification key
M01	`binance_primary_4h`
M02	`binance_matched_4h`
M03	`binance_matched_1h`
M04	`binance_matched_1d`
M05	`bitstamp_primary_4h`
M06	`bitstamp_matched_4h`

G Simple-control comparisons

Table 12: Base-cost adaptive-minus-control contrasts. All intervals are exploratory, unadjusted, 30-day moving-block summaries; differences are annual log units in percent.

Case	Control	Difference (%)	Interval (%)
M01	Ensemble	2.67	[-2.04, 8.21]
M01	Cash	1.43	[-14.96, 20.12]
M01	Buy and hold	8.06	[-39.06, 68.83]
M01	Volatility target	5.04	[-18.53, 35.97]
M01	Slow resizing	-1.19	[-2.67, -0.02]
M01	Fixed EMA 25/100	4.37	[-7.47, 14.88]
M01	Fixed EMA 100/400	4.06	[-5.47, 17.45]
M02	Ensemble	1.50	[-7.75, 9.69]
M02	Cash	10.14	[-17.94, 44.50]
M02	Buy and hold	-22.07	[-90.65, 68.08]
M02	Volatility target	-8.70	[-45.72, 37.94]
M02	Slow resizing	-0.93	[-2.50, 0.19]
M02	Fixed EMA 25/100	-0.49	[-25.06, 19.60]
M02	Fixed EMA 100/400	4.42	[-12.78, 24.20]
M03	Ensemble	-8.10	[-32.81, 8.40]
M03	Cash	1.06	[-21.19, 26.98]
M03	Buy and hold	-31.15	[-115.91, 68.86]
M03	Volatility target	-17.98	[-73.00, 36.45]
M03	Slow resizing	-11.88	[-34.07, 0.00]
M03	Fixed EMA 100/400	-12.68	[-57.53, 17.90]
M03	EMA 400/1600	-3.97	[-19.64, 10.65]
M04	Ensemble	-5.18	[-19.95, 5.53]
M04	Cash	1.10	[-18.35, 25.38]
M04	Buy and hold	-31.11	[-109.02, 67.23]
M04	Volatility target	-14.55	[-64.86, 45.64]
M04	Slow resizing	-7.38	[-24.11, 4.97]
M04	EMA 5/17	-8.02	[-37.19, 7.81]
M04	EMA 17/67	-4.05	[-24.25, 15.18]
M05	Ensemble	1.78	[-3.31, 7.37]
M05	Cash	1.09	[-15.51, 19.99]
M05	Buy and hold	7.66	[-39.96, 65.96]
M05	Volatility target	3.76	[-19.05, 32.05]
M05	Slow resizing	-1.21	[-2.71, -0.04]
M05	Fixed EMA 25/100	2.76	[-9.43, 14.16]
M05	Fixed EMA 100/400	3.01	[-6.69, 16.16]
M06	Ensemble	0.76	[-9.23, 9.38]
M06	Cash	9.57	[-18.51, 44.39]
M06	Buy and hold	-22.56	[-92.18, 67.48]
M06	Volatility target	-9.25	[-44.92, 34.13]
M06	Slow resizing	-0.92	[-2.56, 0.18]
M06	Fixed EMA 25/100	-2.18	[-28.28, 19.50]
M06	Fixed EMA 100/400	3.67	[-13.16, 23.33]

H Post-protocol audit of the subsequently supplied daily bot

The original separate bot became available on 22 October 2026, after the frozen EMA study and its initial manuscript had completed. It was a native daily 40/50/60/80/100/150-day SMA vote with 40% volatility targeting, 30 log returns and a 0.15 weight band. It is not the four-hour adaptive EMA selector studied above. The latter's negative result therefore does not test the original SMA algorithm or verify its supplied performance claims. The original 14-file source archive and its hashes are retained.

The delivered correction, named `legacy_daily`, retains that SMA vote and risk target but requires 150 continuous candles, resets after gaps, checks resizing at executable opening valuation, uses shared entry/full-exit rules, and solves exact post-fee weights without borrowing cash. These corrections differ from a fee-only repair of the old close-trigger loop. The paper EMA strategy remains the operational default for protocol consistency; that choice is not a claim that it maximizes future returns.

On the already inspected 1 January 2025 through 20 October 2026 window, the delivered daily variant returned 11.8784% at assumed 25 bps per side, with 25.1663% maximum drawdown sampled at daily closes. Its 100 bps return was -8.4427%. These are corrected-implementation outcomes, not validation of the old performance claims. Daily-close drawdowns omit intraday excursions and are not the same measurements as four-hour drawdowns. No authenticated broker fills or prospective profitability test are supplied by this comparison.

H.1 A dated exploratory matched-settings contrast

After those outcomes had been inspected, a new protocol was frozen on 23 October. It compares the fixed SMA vote and fixed EMA 9/34 at each target 20% and 40%, using identical native daily rows, 30-return volatility, a 0.10 opening-weight band, 334-bar contiguous feature warm-up, unit exposure cap, prior-close signals, next-open accounting and inherited prewindow positions. The EMA 9/34 pair is the paper's upward clock conversion of 50/200; the 334-bar gate is applied to both signals. These common-setting SMA variants differ from the delivered 0.15-band, 150-bar variant and must not replace its result. Both share the same valid feature history at the scoring boundary; no selector is fitted or ranked.

The eight books are the four signal/target combinations, cash, buy-and-hold and volatility-only controls at both targets. Every book is rerun at 0/10/25/50/100 bps. In the tables, suffixes `20` and `40` denote target percentages; `volatility` denotes the signal-free risk-target control. Tables 13–14 retain the 40 paths and ten within-target cost contrasts; companion CSVs retain all metrics and 15/30/60-day approximate paired intervals from 5,000 draws. Source, protocol, input and result hashes, source ZIP, failed/completed attempts and exact archived-source replay are recorded separately from the registered six market cases. Replaying them creates no new independent observations.

At 25 bps the 20%-target SMA/EMA returns are 7.4930% and -2.4915%; at 40% they are 12.2485% and 3.5831%. The corresponding annualized mean-log contrasts are 5.4077% [-6.3797%,15.2184%] and 4.4566% [-12.6727%,17.7382%] using 30-day blocks. Both intervals include zero. Matching target parameters does not equalize realized risk or exposure, as the controls show; this is a signal contrast at matched settings, not causal alpha. Its post-outcome design, unadjusted intervals and reused history do not establish a winning variant, fresh-holdout confirmation or future profitability. No strategy is selected from these outcomes.

Table 13: Post-outcome matched daily settings. Returns, realized annual volatility, exposure and drawdown are percent; turnover is unscaled. No strategy is selected.

Cost	Book	Return	Vol.	Exposure	Turnover	Drawdown
0	sma_20	11.76	13.47	27.46	15.58	-15.23
0	ema_20	1.30	14.41	27.78	15.25	-20.58
0	sma_40	21.11	22.18	45.99	30.41	-21.31
0	ema_40	9.56	23.87	46.36	22.44	-28.08
0	cash	0.00	0.00	0.00	0.00	0.00
0	buy_hold	-11.27	43.11	100.00	0.00	-52.97
0	volatility_20	-4.59	21.82	52.35	3.96	-35.61
0	volatility_40	-11.24	37.25	89.16	2.10	-50.55
10	sma_20	10.04	13.48	27.46	15.58	-16.13
10	ema_20	-0.24	14.40	27.78	15.24	-21.22
10	sma_40	17.49	22.17	45.99	30.41	-22.81
10	ema_40	7.13	23.85	46.36	22.43	-28.76
10	cash	0.00	0.00	0.00	0.00	0.00
10	buy_hold	-11.27	43.11	100.00	0.00	-52.97
10	volatility_20	-4.96	21.82	52.35	3.95	-35.71
10	volatility_40	-11.43	37.25	89.16	2.10	-50.61
25	sma_20	7.49	13.48	27.46	15.58	-17.46
25	ema_20	-2.49	14.39	27.78	15.23	-22.44
25	sma_40	12.25	22.17	45.99	30.40	-25.00
25	ema_40	3.58	23.83	46.36	22.42	-29.78
25	cash	0.00	0.00	0.00	0.00	0.00
25	buy_hold	-11.27	43.11	100.00	0.00	-52.97
25	volatility_20	-5.53	21.82	52.35	3.95	-35.85
25	volatility_40	-11.71	37.24	89.16	2.10	-50.70
50	sma_20	3.39	13.50	27.46	15.57	-19.72
50	ema_20	-6.14	14.40	27.78	15.22	-24.44
50	sma_40	4.03	22.18	45.99	30.38	-28.62
50	ema_40	-2.07	23.82	46.36	22.39	-31.47
50	cash	0.00	0.00	0.00	0.00	0.00
50	buy_hold	-11.27	43.11	100.00	0.00	-52.97
50	volatility_20	-6.46	21.82	52.35	3.95	-36.09
50	volatility_40	-12.17	37.24	89.16	2.10	-50.84
100	sma_20	-4.36	13.56	27.46	15.56	-24.11
100	ema_20	-13.03	14.48	27.78	15.20	-28.27
100	sma_40	-10.64	22.24	45.99	30.36	-35.38
100	ema_40	-12.46	23.89	46.36	22.34	-36.37
100	cash	0.00	0.00	0.00	0.00	0.00
100	buy_hold	-11.27	43.11	100.00	0.00	-52.97
100	volatility_20	-8.29	21.82	52.35	3.95	-36.56
100	volatility_40	-13.09	37.22	89.16	2.10	-51.13

Table 14: SMA-minus-EMA annual mean-log differences in percent, at matched target settings. Approximate unadjusted 95% 30-day block intervals; all 15/30/60-day intervals are in paired.csv.

Cost	Contrast	Difference	Interval
0	sma_20_minus_ema_20	5.45	[-5.88, 14.95]
0	sma_40_minus_ema_40	5.56	[-10.87, 18.41]
10	sma_20_minus_ema_20	5.44	[-6.07, 15.03]
10	sma_40_minus_ema_40	5.12	[-11.67, 18.14]
25	sma_20_minus_ema_20	5.41	[-6.38, 15.22]
25	sma_40_minus_ema_40	4.46	[-12.67, 17.74]
50	sma_20_minus_ema_20	5.36	[-6.84, 15.42]
50	sma_40_minus_ema_40	3.35	[-14.50, 17.23]
100	sma_20_minus_ema_20	5.27	[-7.92, 16.06]
100	sma_40_minus_ema_40	1.14	[-18.13, 16.39]